Speech Communication 175 (2025) 103317

Contents lists available at ScienceDirect

Speech Communication

journal homepage: www.elsevier.com/locate/specom

Direct speech-to-speech neural machine translation: A survey

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ARTICLE INFO

Keywords:
Direct speech-to-speech translation
Representation learning
Textless training
Discrete units
Pre-training
Self-supervised learning

ABSTRACT

Speech-to-Speech Translation (S2ST) models transform speech from one language to another target language with the same linguistic information. S2ST is important for bridging the communication gap among communities and has diverse applications. In recent years, researchers have introduced direct S2ST models, which have the potential to translate speech without relying on intermediate text generation, have better-decoding latency, and the ability to preserve paralinguistic and non-linguistic features. However, direct S2ST has yet to achieve quality performance for seamless communication and still lags behind the cascade models in terms of performance, especially in real-world translation. To the best of our knowledge, no comprehensive survey is available on the direct S2ST system, which beginners and advanced researchers can look upon for a quick survey. The present work extensively reviews direct S2ST models, data and application issues, and performance metrics. We critically analyze the models' performance over the benchmark datasets and provide research challenges and future directions.

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https://doi.org/10.1016/j.specom.2025.103317
Received 16 December 2024; Received in revised form 25 September 2025; Accepted 10 October 2025
Available online 16 October 2025
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1. Introduction

The S2ST task is the process of transforming speech in a source language to speech in a target language. It finds applications in live audio/video translation, international conferences and meeting translations, educational video translation, and movie dubbing, to name a few. The rising computational power of computing devices, the expanding bandwidth of communication channels, and the progress in deep learning (DL) and machine learning (ML) techniques have enabled seamless communication.

Traditionally, the S2ST problem is solved by the cascade approach in that automatic speech recognition (ASR), machine translation (MT), and text-to-speech synthesis (TTS) models are glued together (Lavie et al., 1997; Nakamura et al., 2006). Another approach to building a cascade S2ST model is by chaining speech-to-text translation (ST) and TTS models (Jia et al., 2019a; Wang et al., 2021c; Dugan et al., 2023). Nonetheless, the cascade models have been the de-facto choice for S2ST for a long time due to the easier availability of the pre-trained component models (ASR, MT, and TTS). However, they are being challenged by direct S2ST models in recent years owing to error propagation, higher training time, and memory cost of cascade models (Lee et al., 2022a).

Direct S2ST models translate source speech into target speech without using intermediate text representation. They are being popularized by the possibility of learning paralinguistic and non-linguistic features (Jia et al., 2019b, 2022b; Lee et al., 2022a). Paralinguistic features carry information regarding accent, dialect, and speakers' attitude, style, gestures, moods (e.g., interested or indifferent), emotions (e.g., happy, sad, or angry), articulation, and identity. Whereas Non-linguistic features are laughter, crying, coughing, sneezing, and yawning (Mittal, 2014; Kadali and Mittal, 2020). By translation of these features, we refer to the preservation or reproduction of such sounds or style across languages—for example, ensuring that laughter, sighs, or crying in the source speech are appropriately represented in the target output. This not only maintains communicative intent but also enhances the naturalness and emotional fidelity of S2ST. Direct S2ST models do not require an intermediate text representation, enabling these models to be used for languages that lack a writing system (i.e., languages that are primarily spoken rather than written). Such languages, often called unwritten languages, constitute 40% of all languages (Lee et al., 2022b). Text-based model training is not feasible for these languages. Numerous efforts have been made to develop multilingual ASR systems (Chung et al., 2021; Babu et al., 2022; Radford et al., 2023; Pratap et al., 2024), which significantly enhance the performance of speech models for low-resource languages, such as ASR and ST. Some of these models (Hsu et al., 2021; Babu et al., 2022) also lay the groundwork for building S2ST models for unwritten languages. Parallel efforts have focused on designing zero-shot and zero-resource-constrained speech models (Nabhani et al., 2024; Khurana et al., 2024), paving the way for the development of direct S2ST models under zero-resource or zero-shot settings (Zhang et al., 2023; Rubenstein et al., 2023; Nguyen and Sakti, 2025).

Direct models generally follow an end-to-end (E2E) training approach, reducing error propagation and offering lower latency than cascade models (Jia et al., 2019b, 2022b; Lee et al., 2022a). However, despite their advantages, these models encounter notable challenges such as: (a) getting sufficient parallel speech corpus in two different languages is extremely hard, thus hampering model development, (b) training with the speech of unwritten languages and their evaluation, (c) the potential threat of voice cloning, (d) absence of metrics directly taking generated and reference speech as input and returning a quality score, and (e) segmentation issue especially in simultaneous S2ST, and so on. Further advancements are still required for S2ST systems to attain the level of quality necessary for hassle-free communication.

To provide an overview of research done in the S2ST field and a bucket of open problems, the present work comprehensively reviews the direct S2ST literature, whose broad overview is shown in Fig. 1. In direct S2ST literature, the terms "direct"and "E2E"are often used interchangeably. The term *E2E* has various definitions. The one most relevant to the direct S2ST task is based on *joint training* and is defined as: "In terms of model training, E2E suggests estimating all parameters of all components of a model jointly using a single objective function that is consistent with the task at hand (Prabhavalkar et al., 2024)". Similarly, in the literature on direct ST, the terms *E2E* and *Direct* are generally used interchangeably, and it is also consistent as per the E2E definition. However, in the case of direct S2ST, exceptions exist. Some models are referred to as *Direct* even though they do not follow a fully unified (E2E) training approach (Diwan et al., 2024; Kaur et al., 2024; Dong et al., 2024). These models are considered direct because they avoid the use of intermediate text representations. In our literature survey, we classify all S2ST models that do not rely on intermediate text representations as *Direct*. The use of intermediate *discrete unit* representations is allowed, as these units are assumed to encode both linguistic and paralinguistic features to some extent. This opens up the possibility of transferring paralinguistic cues such as speaking style and emotion. Another advantage of such models is their potential applicability to *unwritten languages*—that is, languages that are primarily spoken but not written. Conversely, we categorize models that utilize intermediate text representations as *cascade S2ST* systems. Similarly, this paper uses these terms interchangeably in the same context. There are various ways to characterize direct S2ST models, but no single approach is universally applicable to all model architectures. In this work, we categorized S2ST models from three different perspectives: framework, application, and size, as depicted in Fig. 1.

Overall, the manuscript is organized as follows: Section 2 formally defines the S2ST task; Section 3 discusses the set of techniques to address data scarcity. The performance metrics are presented in Section 4 while Section 5 elucidates the segmentation and representation issues. S2ST models are classified and presented in Section 6.1.2, while their

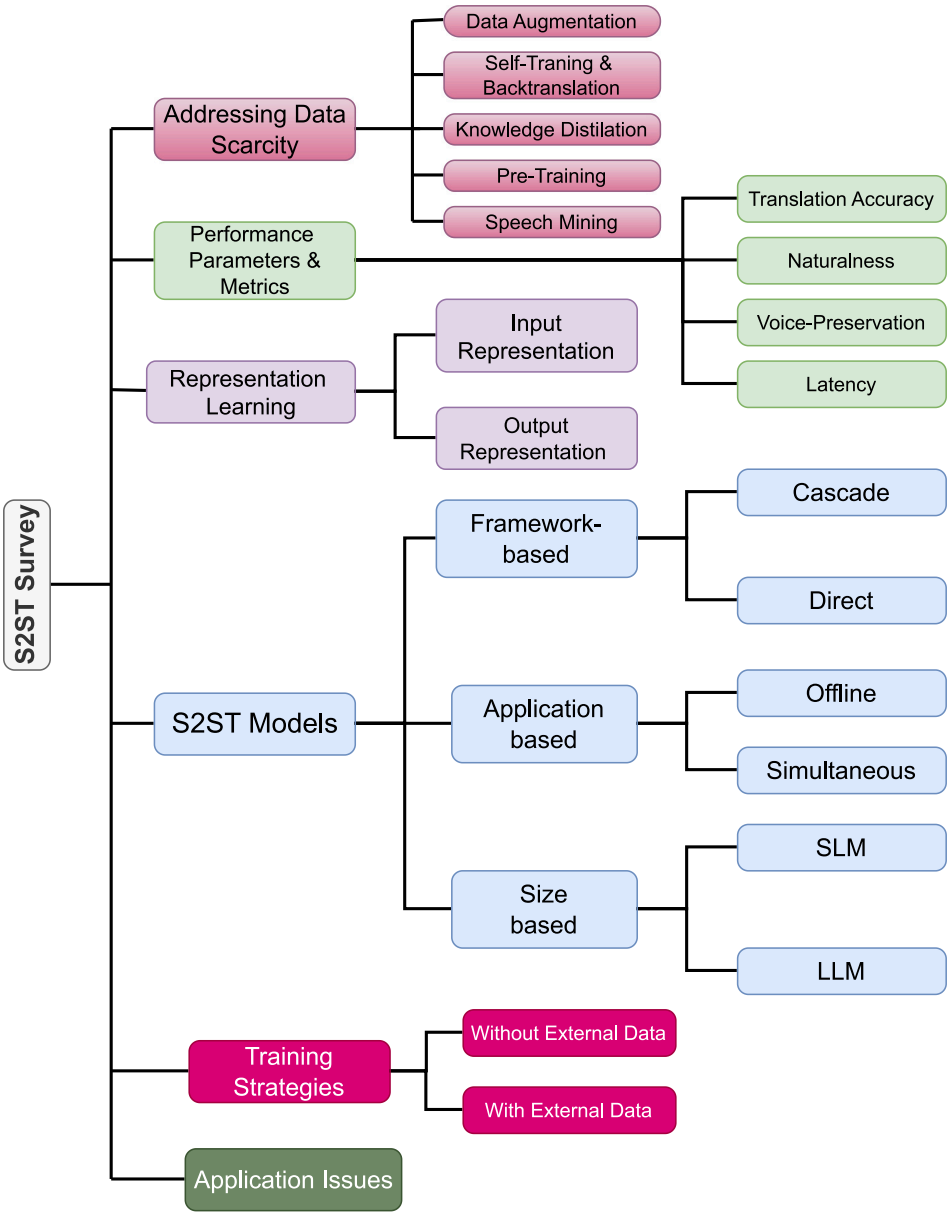


Fig. 1. Overview of direct S2ST literature review.

training strategies are discussed in Section 7. The experimental results are presented in Section 8, followed by a discussion of application issues in Section 9. The key challenges are outlined in Section 10, and the paper is concluded in Section 11.

2. Task definition

Direct S2ST is a process that involves mapping the sequence of source language speech frames to corresponding target language speech frames while preserving the same linguistic information without intermediate text generation. Formally, given a parallel speech corpus, denoted as $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{n}$, where n is the total number of parallel speech pairs, and i indexes each pair. In this context, $x = \{f_1^s, \ldots, f_K^s\}$ and $y = \{f_1^t, \ldots, f_L^t\}$ represent the source and target speech utterances, respectively. Here, f^s and f^t refer to the source and target speech frames, and K and L represent utterance length in frames, respectively. The objective of direct S2ST models is to maximize the conditional probability $\mathcal{P}(y|x;\theta)$ as in (1) or minimize the negative log-likelihood loss (2).

$$\mathcal{P}(y|x;\theta) = \prod_{t=1}^{L} \mathcal{P}(f_t^t | f_{<t}^t, x; \theta) \tag{1}$$

$$\mathcal{L}_{(x,y)\in\mathcal{D}} = -\sum_{i=1}^{n} \log \mathcal{P}(y_i|x_i;\theta) \tag{2}$$

where $\mathcal{L}_{(x,y)\in\mathcal{D}}$ is the cumulative loss on the dataset $\mathcal{D}$, θ is a model parameter, and t is time stamp when f_t^t is being predicted. Note that the problem formulation given in (1) is for *Autoregressive* (AR) models.[1]

On the other hand, cascade models have access to source and target transcripts that correspond to parallel speech data $\mathcal{D}$ denoted as $\mathcal{C} = \{(\mathcal{U}_i, \mathcal{V}_i)\}_{i=1}^{n}$, where $\mathcal{U} = \{u_1, \ldots, u_K\}$ and $\mathcal{V} = \{v_1, \ldots, v_L\}$ represent

[1] Non-autoregressive (NAR) models are an alternative modeling approach that has been proposed in the past few years. Only a sparse number of works exist in the literature on S2ST where they use NAR in decoders of encoder–decoder frameworks. We discuss NAR briefly in Section 6.2.1.

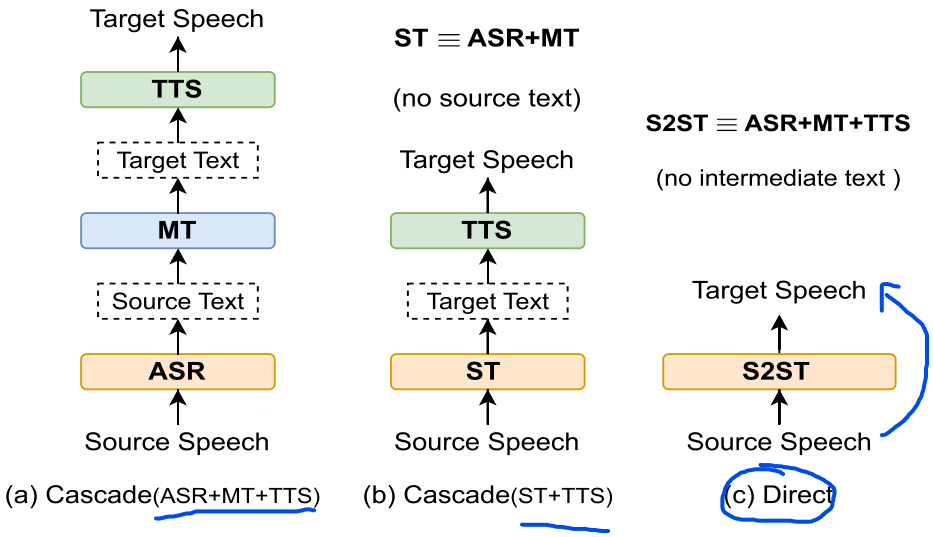


Fig. 2. S2ST model design techniques (a) 3-stage cascade, (b) 2-stage cascade, and (c) direct.

sequences of tokens (words) corresponding to the source and target transcripts, respectively. If the speech dataset also includes transcripts, it can be expressed as $\mathcal{D} \cup \mathcal{C} = \{(x_i, \mathcal{U}_i, y_i, \mathcal{V}_i)\}_{i=1}^{n}$.

Similar to direct S2ST, the objective of direct ST can be defined as follows: the goal is to translate source speech x into a target language transcript $\mathcal{V}$, where v_t represents the tth target token in the target language. Given a speech-text parallel dataset $\mathcal{D}_{\text{ST}} = \{(x_i, \mathcal{V}_i)\}_{i=1}^{n}$, the objective is to maximize the conditional probability $\mathcal{P}(\mathcal{V} \mid x; \theta^{st})$, or equivalently, to minimize the negative log-likelihood loss:

$$\mathcal{P}(\mathcal{V}|x;\theta^{st}) = \prod_{t=1}^{L} \mathcal{P}(v_t|v_{<t}, x; \theta^{st}) \tag{3}$$

$$\mathcal{L}^{\text{ST}}_{(x,\mathcal{V})\in\mathcal{D}_{\text{ST}}} = -\sum_{i=1}^{n} \log \mathcal{P}(\mathcal{V}_i|x_i;\theta^{st}) \tag{4}$$

where $\mathcal{L}^{\text{ST}}$ is the cumulative loss for the direct ST model on dataset $\mathcal{D}_{\text{ST}}$, and θ^{st} represents the model parameters. Eq. (3) corresponds to the autoregressive formulation of the model, where the target token v_t is predicted given all previous tokens $v_{<t}$ and the source speech input x.

In the 3-stage cascade model, the optimization objectives are as follows:

$$\mathcal{L}_{asr} = -\sum_{i=1}^{n} \log \mathcal{P}(\mathcal{U}_i|x_i;\theta^{asr}) \tag{5}$$

$$\mathcal{L}_{mt} = -\sum_{i=1}^{n} \log \mathcal{P}(\mathcal{V}_i|\mathcal{U}_i;\theta^{mt}) \tag{6}$$

$$\mathcal{L}_{tts} = -\sum_{i=1}^{n} \log \mathcal{P}(y_i|\mathcal{V}_i;\theta^{tts}) \tag{7}$$

A 3-stage cascade model is built by independently minimizing the losses in (5), (6), and (7). A direct ST model (Bérard et al., 2018; Kano et al., 2020) optimizes the loss in (4) (may also use losses in (5), (6) in a multitask setup). This ST model is combined with the TTS model to form a 2-stage cascade S2ST model.

3. Strategies for addressing data scarcity

Parallel speech corpora are crucial for training direct S2ST models, and their scarcity can significantly affect their training, performance, and robustness. Creating parallel speech corpora requires substantial resource investment (Chen et al., 2023b), resulting in limited available datasets, as shown in Table 1. While this scarcity of supervised S2ST datasets prevents the application of direct S2ST systems to low-resource languages, techniques like data augmentation, pre-training, self-training, back-translation, knowledge distillation, and speech mining are employed to address data scarcity and enable S2ST for low-resource languages. These methods are further elaborated in the following sections.

3.1. Data augmentation

Data augmentation (DA) refers to techniques that boost the number of data points to enhance diversity without requiring the explicit collection of real supervised data points (Feng et al., 2021). The DA techniques suitable for various NLP tasks may not be directly applicable to S2ST since S2ST involves parallel speech, posing a challenge for augmenting existing data. Speech data can be augmented in various ways. For example, by adding noise, speed and pitch perturbation, time and frequency masking, etc. Formally, for a given dataset $\mathcal{D}$ as defined in Section 2, a transformation function $\mathcal{F}_{tf}$ is applied to the original input speech x to create a new speech sample $x^{(j)}$ with distorted acoustic features. The corresponding target speech y remains unchanged, resulting in a transformed dataset:

$$\mathcal{D}_{tf} = \{(x_i^{(j)}, y_i) \mid x_i^{(j)} = \mathcal{F}_{tf}^{(j)}(x_i),\ 1 \le j \le n_{tf},\ 1 \le i \le n\} \tag{8}$$

where n_{tf} denotes the number of transformations applied to each input sample. Additionally, to leverage unsupervised data, a pseudo-labeled dataset $\mathcal{D}_{pseu}$ can be generated using pre-trained MT and TTS models. Finally, the original dataset, transformed dataset, and pseudo-labeled dataset are combined to form the augmented dataset: $\mathcal{D}_{aug} = \mathcal{D} \cup \mathcal{D}_{tf} \cup \mathcal{D}_{pseu}$. This augmentation strategy increases both the size and diversity of the dataset, enhancing the model's ability to generalize.

In Jia et al. (2022b), the DA technique *ConcatAug* is employed to preserve the speaker's voice on their turn. In particular, training data is enhanced by randomly selecting pairs of training examples and concatenating their source speech, target speech, and target phoneme sequences to create new training examples. These new examples incorporate the voices of two different speakers in both the source and target speech, allowing the model to learn from instances that include speaker turns. Many studies utilize synthesized speech from the target text in ST data (Jia et al., 2019b, 2022b; Lee et al., 2022a; Nachmani et al., 2024), while others make use of MT and TTS models to label ASR corpora through pseudo-labeling (Dong et al., 2022; Jia et al., 2022a; Popuri et al., 2022; Huang et al., 2023b).

3.2. Pre-training

Pre-training (Popuri et al., 2022; Wei et al., 2023) aims to improve the performance of the task in one domain with the learned knowledge

Table 1
Details of the available S2ST dataset (EP: European Parliament, CV: Common Voice). The domain indicates the source of the raw data, while the nature specifies whether the speech is spontaneous, read, or synthesized. "spontaneous"means both sides of the speech are spontaneous, whereas "Read/Synthesized"means one side is read-speech and the other is synthesized speech. (STC (Shimizu et al., 2014), MaSS (Zanon Boito et al., 2020), VoxPopuli (Wang et al., 2021a), CVSS-C/T (Jia et al., 2022c), SpeechMatrix (Duquenne et al., 2023), LibriS2S (Jeuris and Niehues, 2022), FLEURES (Conneau et al., 2022), Hi-EnS2S (Gupta et al., 2025a)).

Dataset	Domain	Src Lang.	Tgt Lang.	#Hours	#Speakers	Gender
STC	TED Talks, Spont	Jp, En	Jp, En	31	3	Mixed
MaSS	Bible, Read	15 Lang.	15 Lang.	150	80	Mixed
VoxPopuli	EP, Spont	15 Lang.	15 Lang.	17.3K	80	Mixed
CVSS-C	CV, Read/Synth	21 Lang.	English	719	1 Canon.	F
CVSS-T	CV, Read/Synth	21 Lang.	English	784	Cloned	M/F
SpeechMatrix	EP, Spont	17 Lang.	17 Lang.	418k	50	Mixed
LibriS2S	Librivox, Read	Geerman	English	52.5	42	Mixed
FLEURES	Wikipedia, Read	102 Lang.	102 Lang	1.4k	✗	Mixed
Hi-EnS2S	TED, Spont/Synth	Hi, En	Hi, En	116	1 Canon.	F

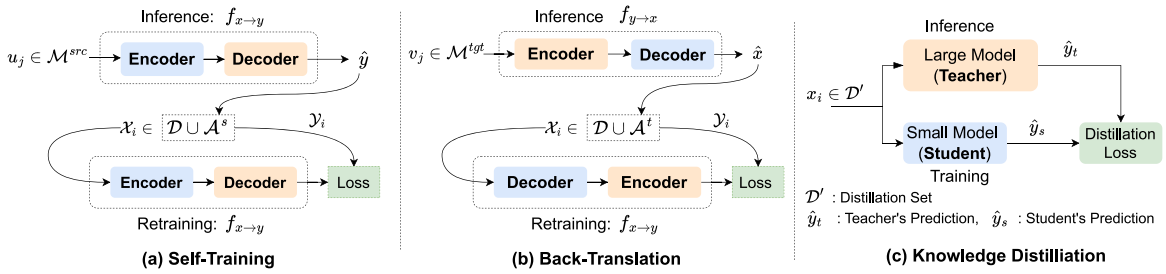


Fig. 3. Strategies for addressing data scarcity.

from the tasks in another domain (Pan and Yang, 2010; Wang et al., 2023). Models learn salient features from an extensive training corpus during pre-training. Subsequently, the pre-trained model is applied to another (target) task, employing smaller datasets for fine-tuning (Oord et al., 2018; Tang et al., 2022). Formally, given a model $\mathcal{M}_{pt}$ parameterized by θ, the pre-training process involves optimizing θ to minimize a pretraining loss function $\mathcal{L}_{pt}$ over a large dataset $\mathcal{D}_{pt}$:

$$\theta^* = \arg\min_{\theta} \mathcal{L}_{pt}(\mathcal{M}_{pt}(x;\theta), \bar{y}), \quad \forall (x, \bar{y}) \in \mathcal{D}_{pt} \tag{9}$$

Here, $\bar{y}$ represents the target, which may or may not align with the downstream task. The pre-training process can be supervised, semi-supervised, or unsupervised, depending on the task requirements and the availability of labeled data. Consequently, the formulation of (9), illustrated for supervised pre-training, may vary accordingly. Following pre-training, the model $\mathcal{M}_{pt}$ is adapted to the downstream task by adding task-specific layers, forming $\mathcal{M}'_{pt}$. The model is then fine-tuned on a fine-tuning dataset $\mathcal{D}_{ft}$:

$$\theta^* = \arg\min_{\theta} \mathcal{L}_{ft}(\mathcal{M}'_{pt}(x;\theta), y), \quad \forall (x, y) \in \mathcal{D}_{ft} \tag{10}$$

where $\mathcal{L}_{ft}$ represents the fine-tuning loss function, and y corresponds to the target labels in the downstream task.

In recent works, self-supervised strategies are used for pre-training, to leverage the availability of large unlabeled datasets (Popuri et al., 2022; Wei et al., 2023). Several pre-trained DL models, including Wav2Vec2.0 (Baevski et al., 2020), HuBERT (Hsu et al., 2021), w2v-BERT (Chung et al., 2021), WavLM (Chen et al., 2022), unit-mBART trained on speech-units (Popuri et al., 2022), and VQ-VAE (Oord et al., 2017), which are trained on extensive unlabeled data, serve as the foundation for various NLP and speech tasks. The pre-trained models are used as encoders/decoders in direct S2ST models (Jia et al., 2019b, 2022b; Popuri et al., 2022; Lee et al., 2022c) (refer to Section 6.1.2)

3.3. Self-training and back-translation

Self-training and Back-translation (BT) algorithms leverage monolingual data (text or speech, depending on the model's nature) to train models requiring supervised data but lacking a supervised parallel corpus. These algorithms use the model's confident predictions for monolingual data, with the model initially trained on available parallel data (Edunov et al., 2018; Yang et al., 2022). Self-training utilizes source monolingual data, while BT utilizes target monolingual data to generate augmented data. Consider an utterance-level parallel speech corpus $\mathcal{D}$, as explained in Section 2. Additionally, we have source monolingual corpus $\mathcal{M}_{src} = \{u_j\}_{j=1}^{m}$, and target monolingual corpus $\mathcal{M}_{tgt} = \{v_k\}_{k=1}^{q}$, where $m, q \gg n$. The forward translation model $f_{x\to y}$ is optimized for the parameter $\theta_{x\to y} = \text{argmin}_\theta \sum_{(x_i,y_i)\in\mathcal{D}} \mathcal{L}(f^{\theta}_{x\to y}(x_i), y_i)$, where $\mathcal{L}$ is the loss function. Similarly, another model in backward direction, $f_{y\to x}$, is optimized for the parameter $\theta_{y\to x} = \text{argmin}_\theta \sum_{(x_i,y_i)\in\mathcal{D}} \mathcal{L}(f^{\theta}_{y\to x}(y_i), x_i)$. Using the already trained model $f_{x\to y}$ and $\mathcal{M}_{src}$, pseudo-labeled utterances $\hat{y}$ are generated to create a new auxiliary parallel corpus $\mathcal{A}^s = \{(u_j, \hat{y}_j)\}_{j=1}^{m'}$ by selecting confident predictions. In self-training, as shown in Fig. 3(a), the model $f_{x\to y}$ is retrained on the augmented data $\mathcal{D} \cup \mathcal{A}^s$ (He et al., 2020). Similarly, an auxiliary parallel corpus $\mathcal{A}^t = \{(\hat{x}_k, v_k)\}_{k=1}^{q'}$ is generated using the backward-trained model $f_{y\to x}$. When employing back-translation (BT) the model $f_{x\to y}$ is trained in the forward direction on the newly augmented data $\mathcal{D} \cup \mathcal{A}^t$ as depicted in Fig. 3(b) (Sennrich et al., 2016). Some studies utilize a denoising version of BT, where noise is added to the input (Fu et al., 2023). Several studies demonstrate that self-training (Pino et al., 2020) and BT (Pino et al., 2020) are highly beneficial, particularly for low-resource languages compared to high-resource ones. These algorithms also benefit S2ST directly or indirectly, with the indirect method proving more effective (Popuri et al., 2022; Nachmani et al., 2024).

3.4. Knowledge Distillation

Knowledge Distillation (KD) transfers learned knowledge from a large ensemble model to a smaller model as shown in Fig. 3(c) (Hinton et al., 2015; Treviso et al., 2023). The KD process is based on the teacher–student learning paradigm: the larger model serves as the teacher, while the smaller model acts as the student. As illustrated

in Fig. 3(c), $\mathcal{D}'$ represents the distillation set used to transfer learned patterns from the teacher model to the student model. The teacher and student model predictions are denoted as $\hat{y}_i^{(t)} = \textbf{Teacher}(x_i)$ and $\hat{y}_i^{(s)} = \textbf{Student}(x_i)$, respectively. The teacher model's predictions are referred to as *soft targets*, while the actual target labels are termed *hard targets*. The total loss combines the *distillation loss* $\mathcal{L}_{distill}(\hat{y}_i^{(s)}, \hat{y}_i^{(t)})$ and the *task-specific loss* $\mathcal{L}_{task}(y_i, \hat{y}_i^{(s)})$, formulated as:

$$\mathcal{L}_{total} = \alpha \mathcal{L}_{distill}(\hat{y}_i^{(s)}, \hat{y}_i^{(t)}) + (1-\alpha)\mathcal{L}_{task}(y_i, \hat{y}_i^{(s)}) \tag{11}$$

Here, the parameter α controls the balance between the contributions of the soft targets and the hard targets. When the distillation set consists of unsupervised data, $\alpha = 1$, and only the distillation loss is optimized. The soft targets generated by the teacher model provide a richer and more generalized representation compared to hard targets, enabling the student model to achieve better generalization (Hinton et al., 2015)

KD has demonstrated its utility in resource-constrained tasks such as ASR (Hinton et al., 2015), ST (Liu et al., 2019; Inaguma et al., 2021), and S2ST (Huang et al., 2023c). KD can be applied to distill knowledge into specific components of an S2ST model – such as the encoder, decoder, or vocoder (Le et al., 2024) – or the entire model (Ma et al., 2020a). For instance, (Chen et al., 2023c) distilled knowledge from a text-based model into unit-MBART, a speech model. Additionally, KD has been employed to transfer knowledge from offline direct S2ST models to Simul-S2ST systems (Ma et al., 2020a). Similarly, KD has been used to enhance the performance of NAR direct S2ST models by leveraging knowledge from autoregressive (AR) direct S2ST models (Huang et al., 2023c; Fang et al., 2024a). In particular, Huang et al. (2023c) applies cross-modal distillation from audio-only pre-trained S2ST (Popuri et al., 2022) to audio–visual models. Doing so initializes the audio encoder and unit decoder and alleviates the low-resource problem of audio–visual data. Similarly, by leveraging high-resource language models, KD can be effective for low-resource unwritten languages.

3.5. Speech mining

Speech mining is an extension of **bitext mining** (Resnik, 1998), designed to discover parallel speech corpora from monolingual speech corpora. Bitext refers to text-to-text parallel data where the source and target sentences are in different languages. Bitext mining has been done in a supervised and unsupervised way (no cross-lingual resources like parallel text or bilingual lexicons are used) (Keung et al., 2020). Multilingual fixed-length sentence embedding (Heffernan et al., 2022) using KD, contrastive learning (Tan and Koehn, 2022) are used for mining bitext in low-resource settings. Sentence embedding serves to cluster similar sentences closely together within the latent space (Schwenk, 2018; Artetxe and Schwenk, 2019b). Similarly, a multilingual speech encoder is utilized to transform variable-length speech utterances into fixed-length embedding vectors for speech mining. Speech mining has been explored in several works, such as Duquenne et al. (2021), which employed a teacher–student approach to training a multilingual speech encoder. This encoder generates speech representations that align with the outputs of a text embedding layer. Margin-based similarity metrics are applied to extract parallel speech corpora (Artetxe and Schwenk, 2019a; Duquenne et al., 2023). The similarity margin is calculated as:

$$\text{Sim}(x, y) = \cos(\mathcal{E}_x, \mathcal{E}_y) - \left(\sum_{\mathcal{E}_z \in \text{NN}_k(\mathcal{E}_x)} \frac{\cos(\mathcal{E}_x, \mathcal{E}_z)}{2k} + \sum_{\mathcal{E}_z \in \text{NN}_k(\mathcal{E}_y)} \frac{\cos(\mathcal{E}_y, \mathcal{E}_z)}{2k} \right) \tag{12}$$

Here, x and y represent the source and target speech segments, while $\mathcal{E}_x$ and $\mathcal{E}_y$ denote their corresponding speech embeddings in the source and target languages, respectively. $\text{NN}_k(\mathcal{E}_x)$ represents the k-nearest neighbors of x, and $\text{NN}_k(\mathcal{E}_y)$ similarly refers to the k-nearest neighbors of y. Margin-based similarity has demonstrated superior performance compared to the simple cosine similarity-based mining approach (Artetxe and Schwenk, 2019a). Several studies have leveraged speech mining to enhance speech corpora, thereby improving model training (Duquenne et al., 2023; Barrault et al., 2023b).

4. Performance parameters & metrics

Evaluating model performance is critical to developing any model for practical applications. Direct S2ST models aim to translate speech from one language to another, requiring various performance parameters to meet acceptable standards. Key parameters include translation quality, speech quality, voice preservation, and latency. Some parameters, like translation quality and speech quality, are essential across all applications, while others, such as voice preservation and latency, vary based on specific requirements. For instance, voice preservation is essential in movie dubbing, whereas low latency is non-negotiable in real-time applications such as Simul-S2ST. In general, for an ideal model, voice preservation (style and emotion) is desirable in all applications, while low latency is preferred in most. However, due to the limitation of the dataset and system design, existing works found it difficult to focus on these parameters simultaneously. Below, we provide a brief explanation of each parameter.

4.1. Translation quality

The quality of translated speech is measured based on its accuracy, fluency, and cultural appropriateness, ensuring that it effectively conveys the intended meaning in the target language (Colina, 2008). In the literature, the translation quality of MT systems are evaluated using various text-based metrics, including Bilingual Evaluation Understudy(BLEU) (Papineni et al., 2002), SacreBLEU (Post, 2018), chrF (Popović, 2015, 2017), BLEURT (Sellam et al., 2020), COMET (Rei et al., 2020, 2022), and BERT-SCORE (Zhang et al., 2019). Currently, this field lacks speech-based standardized quality metrics for evaluating direct S2ST models. To address this gap, the predicted (hypothesized) speech is transcribed into text using a pre-trained ASR model, and text-based evaluation metrics, such as those previously mentioned, are applied. Direct S2ST models (Le et al., 2020; Jia et al., 2019b; Popuri et al., 2022) are frequently evaluated using the BLEU score on the transcribed speech, commonly called ASR-BLEU. The BLEU metric uses n-gram precision and penalizes shorter candidate sequences than the reference. For direct S2ST, 4-grams (where words are tokens) are typically used for comparison. BLEU is formally defined as follows:

$$\text{BP} = \begin{cases} 1 & \text{if ref_len} \leq \text{can_len} \\ \exp\left(1 - \frac{\text{ref_len})}{\text{can_len}}\right) & \text{otherwise} \end{cases} \tag{13}$$

$$\text{BLEU} = \text{BP} \times \exp\left(\sum_{n=1}^{N} w_n \cdot \log(\mathcal{P}_n) \right) \tag{14}$$

$$\mathcal{P}_n = \frac{\#\text{correct n-grams}}{\#\text{total n-grams}} \tag{15}$$

Where BP is the brevity penalty, $\mathcal{N}$ is the maximum n-gram order considered, $\mathcal{P}_n$ is n-gram precision, w_n is the weight for the n-gram precision, *ref_len* is the length of the reference, and *can_len* is the length of the candidate.

These text-based metrics depend on ASR models' performance, which presents challenges for low-resource and unwritten languages. This is primarily due to either the unavailability of ASR systems or the presence of low-quality models (Salesky et al., 2021). Additionally, these metrics are less effective for morphologically rich languages, such as Hindi, where the same word can be expressed in multiple forms.

Recently, Chen et al. (2023a) introduced a metric called $BLASER^2$ is a text-free S2ST quality metric with both unsupervised ($BLASER_u$) and supervised ($BLASER_s$) versions. The unsupervised version, $BLASER_u$, takes the source, reference, and generated speech, embeds them into a common representation space via a shared encoder, and computes a score directly using the following equation:

$$BLASER_u = \frac{cos(h_{src}, h_{mt}) + cos(h_{mt}, h_{ref})}{2} \tag{16}$$

where cos(·, ·) is the cosine similarity function, h_{src}, h_{mt}, and h_{ref} are the source, generated, and reference speech representations, respectively.

The supervised version, $BLASER_s$, requires human-annotated data to train a neural network-based regressor to estimate scores correlating with human-annotated ratings. The following equation shows how $BLASER_s$ is estimated using the neural network regressor:

$$\begin{aligned} BLASER_s = \text{nnr}([&h_{\text{ref}}; h_{\text{mt}}; h_{\text{src}} \odot h_{\text{mt}}; \\ &|h_{\text{src}} - h_{\text{mt}}|; h_{\text{ref}} \odot h_{\text{mt}}; \\ &|h_{\text{ref}} - h_{\text{mt}}|]) \end{aligned}$$

where nnr(·) denotes a two-layer neural network regressor, and [·; ·] indicates vector concatenation. Except for the LASER encoders, the entire model is trained using the Mean Square Error loss (MSE). The current supervised BLASER model is trained on 7 language pairs. However, it is recommended that a broader range of low-resource languages be incorporated to enhance the robustness of the supervised BLASER model. Additionally, there is a need to develop reference-free evaluation metrics, as existing references may only sometimes be of high quality.

4.2. Speech quality

Speech quality refers to the subjective perception of how well a listener understands and interprets spoken language. It encompasses various factors contributing to the auditory experience, including naturalness, clarity, and intelligibility. Naturalness, in particular, gauges the alignment of predicted speech with natural speech, emphasizing factors such as pauses, stress, and inflexion in the voice. However, clarity specifically targets mumbling and unclear phrases, while intelligibility assesses how well the listeners understand the spoken words in the speech. A human-based subjective measure called *Mean Opinion Score (MOS)* is widely used to evaluate predicted speech, as automated metrics are not readily available. It is a numerical measure that represents the average subjective rating given by human listeners to the quality of a speech signal. MOS provides a simple and intuitive measure of speech quality that correlates well with human perception.

Though MOS is widely used for speech quality, it has a few limitations, such as subjective evaluation, bias, limited sample size, time, and cost. It is recommended to report some *objective* metrics alongside MOS, particularly those that effectively capture the perception of speech quality. Examples include the Perceptual Evaluation of Speech Quality (PESQ) (Rix et al., 2001), its extension Perceptual Objective Listening Quality Analysis (POLQA) (Beerends et al., 2013), and Short-Time Objective Intelligibility (STOI) (Taal et al., 2011). These objective algorithms are specifically designed to predict the perceived quality of speech as experienced by human listeners. Additionally, UTMOS (Saeki et al., 2022), DNMOS (Reddy et al., 2021), DNMOS_Pro (Cumlin et al., 2024), and LE-SSL-mos (Qi et al., 2023) introduces a subjective perceptual quality metric through an ensemble model that integrates SSL, ML, and DL techniques to predict the MOS of speech in English and Chinese.

[2] Balanced Accuracy for Speech Evaluation by Recognition.

4.3. Voice preservation

Voice preservation measures the extent to which predicted speech resembles a particular speaker's voice. It can be assessed using metrics designed for evaluating voice cloning. Measuring voice cloning involves a combination of subjective and objective methods to evaluate various aspects of the cloned voice, such as its similarity to the original voice (e.g., through speaker embedding similarity, as used in Dong et al. (2024)), naturalness, intelligibility, and acoustic properties. Several studies (Jia et al., 2019b, 2022b) use MOS-similarity to subjectively evaluate the similarity of the cloned voice to the original, as discussed in Section 4.2.

4.4. Latency

Simultaneous S2ST (Simul-S2ST) systems are evaluated using quality-based metrics for offline S2ST models along with **latency** metrics. Calculating the latency of the offline S2ST is straightforward: It is equal to the time elapsed in starting to produce the output from the decoder. For Simul-S2ST, calculating the average latency poses a significant challenge since it sees only the partial input. In the absence of specific metrics for the Simul-S2ST task, Simul-T2T latency metrics such as Average Proportion (AP) (Cho and Esipova, 2016), Average Lagging (AL) (Ma et al., 2019), and Length-Adaptive Average Lagging (LAAL) have been adopted as proxies. Some of these metrics, which have been further extended for Simul-ST and S2ST, are formally defined as follows:

Average lagging (AL). AL was initially introduced for latency calculation in Simul-T2T systems (Ma et al., 2019) and later adapted for Simul-ST systems (Ma et al., 2020b). Given an input audio sequence $x = \{f_1^s, \ldots, f_k^s\}$ and corresponding predictions $v = \{w_1^t, \ldots, w_l^t\}$ with timestamps $d(w_j)$ for each predicted token, AL is defined as:

$$\text{AL} = \frac{1}{\tau(T_x)} \sum_{i=1}^{\tau(T_x)} d(w_i^t) - \frac{i-1}{|v|} T_x \tag{17}$$

Here, T_x represents the first index at which the full source sentence is utilized, and $|v|$ denotes the length of the target reference. To compute the timestamps $d(w_j^t)$ for each token, (Ma et al., 2021) employs ASR-transcribed text and a forced aligner. AL is non-differentiable due to the use of the min operation. A differentiable variant, Differentiable Average Lagging (DAL), was introduced in Cherry and Foster (2019) for Simul-T2T latency.

Length adaptive average lagging (LAAL). The AL latency metric is prone to *over-generation* scenarios, underestimating latency when the predicted sequence length ($|v^*|$) is shorter than the reference sequence length ($|y|$). To ensure a fair computation of latency in cases such as over-generation and under-generation, LAAL (Papi et al., 2022) was introduced. It uses $\max\{|v|, |v^*|\}$ instead of $|v|$ as in AL. LAAL is formally defined as follows:

$$\text{LAAL} = \frac{1}{\tau'(T_x)} \sum_{i=1}^{\tau'(T_x)} \left(d(w_i^t) - d_i^* \right) \tag{18}$$

where

$$d_i^* = \frac{(i-1) \cdot T_x}{\max\{|v|, |v^*|\}}$$

and d_i^* represents the oracle normalized delay. Similar to AL, as defined in (17), LAAL also relies on ASR and a speech-text aligner.

Average token delay for S2ST. The Average Token Delay (ATD) (Kano et al., 2024) is a duration-aware latency metric that accounts for token durations, distinguishing it from AL (Ma et al., 2019) and LAAL (Papi et al., 2022), which primarily measures the average number of tokens

processed before translation begins. ATD is inherently robust in handling scenarios involving over- or under-generation and is computation-aware, as it calculates the time delay for each input and output token. ATD is formally defined as follows:

$$\text{ATD} = \frac{1}{|f^t|}\sum_{t=1}^{|f^t|}\left(T(f_t^t) - T(f_{a(t)}^s)\right) \tag{19}$$

$$a(t) = \min(t - d(t), g(t)) \tag{20}$$

$$d(t) = (t-1) - a(t-1) \tag{21}$$

Where $T(f_t^t)$ represents the end time of the output token f_t^t, and $T(f_{a(t)}^s)$ denotes the end time of the corresponding input token $f_{a(t)}^s$. The term $a(t)$ refers to the index of the input token corresponding to the output token f_t^t. Finally, $d(t)$ indicates the length of the preceding translation prefix that exceeds the length of the corresponding input prefix. **Note:** In (19), the superscripts t and s in f_t^t and f_t^s denote the target and source tokens, respectively, while the subscript t represents the token index.

Current practice in S2ST works is that authors report only text-based quality metrics, ignoring objective and subjective metrics as described in this section. It is warranted that S2ST models are evaluated holistically using objective and subjective metrics as well. Further, the development of more effective textless S2ST quality metrics is recommended.

5. Segmentation & representation learning

S2ST models essentially take speech and optionally take text as inputs. Before we can train direct S2ST models, we need to segment the speech and text, followed by how to learn their representations. Handling long speech and text sequences is a challenging task (Kim et al., 2017; Tsiamas et al., 2022). The following section discusses segmentation and representation learning related to speech and text.

5.1. Segmentation learning

Breaking down text into segments is relatively simpler compared to speech segmentation, as it typically relies on robust punctuation marks used by current MT models. However, the difficulty of both text and speech segmentation also depends on the language and its syntactic structure and semantics. During the translation process, there is a segmentation policy that decides where to segment the larger chunk. While building parallel speech data, correct alignment and segmentation are needed to better correlate the source and target speech semantic information. Direct S2ST models necessitate intricate speech segmentation, primarily because of the significant role played by the *out-of-order* word relationships between input and output, alongside the absence of linguistic features. Traditionally, manual segmentation has been the norm for speech processing. However, due to its labor-intensive nature, there is a growing need for segmentation learning. Speech segmentation typically relies on either fixed-length splits, which split the speech at fixed lengths, sometimes randomly (Huang et al., 2023c) or *pause*, which splits the speech on Voice Activity Detection (VAD), as outlined by Sohn et al. (1999). A third method, the *hybrid* approach, integrates both length and linguistic cues for segmentation, as discussed by Potapczyk and Przybysz (2020), Gaido et al. (2021), Tsiamas et al. (2022). Notably, the hybrid approach demonstrates superior performance compared to length and pause-based methods (Gaido et al., 2021). However, there is still a gap in the hybrid and manual approaches to segmentation, and future work may consider paying attention to this.

5.2. Representation learning

The representation of speech is a critical factor in designing direct S2ST models, as speech in two different languages constitutes distinct modalities that may exist in separate representation spaces. The performance of these models heavily relies on how effectively the speech data is utilized. One classical approach to representing speech is to use either the *raw waveform* or *spectrogram* (Jia et al., 2019b; Kano et al., 2021; Kim et al., 2024b; Wang et al., 2024). When using the raw waveform, it is directly fed into the encoder, while in the case of the spectrogram, features such as MFCC are extracted every 20 ms and then passed to the encoder, as discussed in Section 6.2.1. Due to the high variability in speech, direct S2ST models that rely on this representation encounter challenges in learning generalized patterns from the data, which imposes constraints on their performance (Kim et al., 2024a). Moreover, their heavy reliance on supervised datasets significantly limits the applicability of direct S2ST models for low-resource languages, where sufficient investment to obtain parallel speech data is often not feasible (Mohamed et al., 2022).

An alternative approach to speech representation involves using *discrete units*, which leverage *representation learning* to convert continuous speech signals into discrete units akin to words and sub-words in the text domain. Speech representation learning refers to algorithms that extract latent features that encapsulate the essential information required for effective speech modeling (Bengio et al., 2013). Various speech representation learning models use self-supervised learning (SSL), which is a subclass of unsupervised learning that leverages raw waveform to learn speech representation to capture syntactic, low-level acoustics, semantic, and lexical information. Such approaches leverage unlabeled speech data, making it possible to include low-resource and unwritten languages in NLP applications. Earlier representation learning models, such as VQ-VAE (Oord et al., 2017) and XL-VAE (Zhang et al., 2021), variants of the Variational Autoencoder (VAE), employed autoencoding and mask-prediction techniques. Subsequent models, including Wav2vec 2.0 (Schneider et al., 2019; Baevski et al., 2020), Wav2vec-C (Sadhu et al., 2021), and XLS-R (Babu et al., 2022), introduced the use of contrastive loss combined with masking. More recent models, such as HuBERT (Hsu et al., 2021), Discrete-BERT (Baevski and Mohamed, 2020), Data2vec (Baevski et al., 2022), WavLM (Chen et al., 2022), and w2v-BERT (Chung et al., 2021), adopt BERT-like pre-training strategies to learn robust speech representations. These advanced models are pre-trained and evaluated with a focus on pretext tasks, enabling them to learn patterns relevant to downstream tasks. These representation methods are believed to capture higher-level features, such as semantics, rather than focusing on low-level acoustic features like those represented by MFCCs. However, there is ongoing debate regarding the extent to which these models capture semantic versus acoustic features (Choi et al., 2024).

To convert continuous speech into discrete units, the representation models discussed above, combined with k-means clustering algorithms, are utilized. Let Encoder_{SR} represent the speech representation model, and f_c denote the clustering model. The process of speech discretization can be described as follows: $\mathcal{H}_{SR} = \text{Encoder}_{SR}(x) = \{h_1, h_2, \ldots, h_T\}$, $z_i = f_c(h_i), \quad z_i \in \{1, 2, \ldots, K\}$, where K is the total number of clusters, and $z = \{z_1, z_2, \ldots, z_T\}$ represents the sequence of discretized units derived from the source speech (Hsu et al., 2021; Lee et al., 2022a). These discrete units are typically represented using cluster indices instead of actual cluster centers (Hsu et al., 2021; Lee et al., 2022a).

VQ-VAE (Lee et al., 2022b; Li et al., 2023), XL-VAE (Zhang et al., 2021), and HuBERT (Lee et al., 2022b; Wei et al., 2023; Kaur et al., 2024; Diwan et al., 2024; Kim et al., 2024a) are widely used for speech quantization in textless training, particularly for low-resource and unwritten languages, as discussed in Section 7.2. To enhance the performance of S2ST models, several studies have employed pre-trained speech representation models, such as Wav2Vec 2.0 (Popuri et al., 2022), w2v-BERT (Li et al., 2023), and XLS-R (Duquenne et al.,

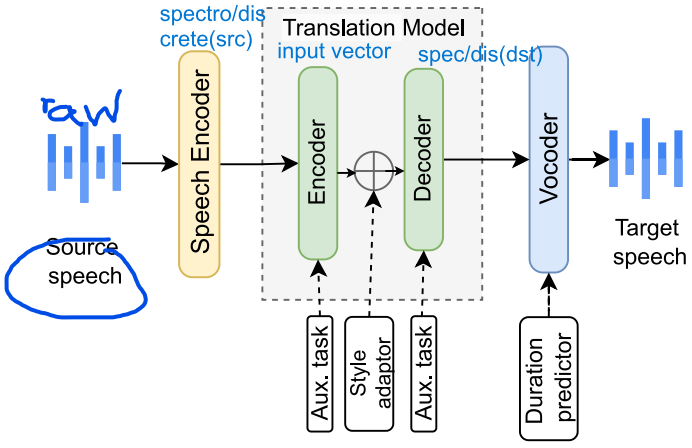


Fig. 4. Typical architecture of offline direct S2ST model.

2023), as encoders for translation models instead of training them from scratch. We not only need better representation learning methods for speech and text but also their joint representation learning. To learn joint speech-text representation where text data is abundant and speech-text pair is limited, *semi-supervised* methods can come in handy. For example, mSLAM (Bapna et al., 2022) combines w2v-BERT and SpanBERT (Joshi et al., 2020) for joint speech-text representation learning, as discussed in Jia et al. (2019a). Similarly, the direct S2ST model proposed in Wei et al. (2023) utilizes HuBERT for speech and text discretization, enabling joint speech-text pretraining and outperforming speech-only models.

6. S2ST models

S2ST models are broadly classified in three different ways, as a single classification approach does not adequately cover all models. These classifications are based on framework, application, and model size, as depicted in Fig. 1. All these subsections logically encompass the full range of S2ST models. However, our focus will remain on direct S2ST models, except for the subsection specifically dedicated to cascade models.

6.1. Framework-based

Based on the framework, S2ST models are categorized into cascade and direct S2ST models. The following subsections discuss each category's fundamental characteristics, advantages, and disadvantages to provide a clear understanding.

6.1.1. Cascade S2ST models

The traditional S2ST systems follow a cascade architecture (Lavie et al., 1997; Ney, 1999; Nakamura et al., 2006; Wahlster, 2013). They are designed either by chaining ASR, MT, and TTS or ST followed by TTS as illustrated in Fig. 2(a) and (b), respectively. These models perform well and are easy to build, as pre-trained ASR, MT, and TTS models are readily available. Either way, the cascade system relies on intermediate text. As such, they face several issues in modeling S2ST effectively. Firstly, they face challenges when dealing with low-resource languages lacking annotated corpora or unwritten languages (Chen et al., 2023c). Secondly, paralinguistic features such as prosody, intonation, emotions, etc. are lost when representing speech via intermediate text, quintessential for building a real-world S2ST system. The intermediate text representation acts as a bottleneck for transferring paralinguistic features. Thirdly, error propagation from one module to another (Jia et al., 2019b; Gupta et al., 2025b), higher training time (Jia et al., 2019b), and memory footprint (due to 3 models vs 1 in direct S2ST) prohibit their application to low-powered devices (Rybakov et al., 2023a). The error propagation analysis in cascade S2ST models conducted by Gupta et al. (2025b) shows a significant decrease in translation accuracy in both directions.

6.1.2. Direct S2ST models

The issues with cascade systems discussed in Section 6.1.1 catapulted the development of direct S2ST systems bypassing the intermediate text generation, reducing training time and memory cost (Lee et al., 2022a). The direct S2ST model is formally defined in Section 2. The ablation study conducted by Lee et al. (2022a) demonstrates that the proposed direct S2ST model requires lower runtime, computational resources, and maximum memory usage compared to cascade models. Direct S2ST models can potentially transfer paralinguistic features of the source speech, such as speaking style, emotion, and rhythm, as they do not rely on intermediate text representations (Jia et al., 2019b). Speaker voice preservation has been achieved through external efforts (Jia et al., 2019b, 2022b). However, the preservation of paralinguistic and non-linguistic features in translated speech has yet to be established without additional efforts.

There has been a lot of work developing direct S2ST models (Jia et al., 2019b, 2022b,c; Huang et al., 2023c; Diwan et al., 2024), etc. Therefore, it is imperative to compare cascade and direct models on quantitative and qualitative metrics (discussed in Section 4). Our literature survey reveals that there is a performance gap between cascade and direct models (both in terms of BLEU and mean opinion score (MOS)) (Jia et al., 2019b, 2022b; Lee et al., 2022b; Zhu et al., 2023; Huang et al., 2023c), although this gap is now narrowing (Chen et al., 2023c; Peng et al., 2024a). These studies, however, are done on limited language pairs and may not generalize. Therefore, it remains to see an exhaustive comparison over multiple and distant language pairs involving large-scale datasets to truly establish the claim that the performance gap is indeed closed. However, despite their advantages over cascade models, direct S2ST models face several challenges, including the difficulty of obtaining sufficient parallel speech corpora in two different languages, which hinders model development; the complexity of training and evaluating models using speech from unwritten languages; the potential risk of voice cloning; the absence of evaluation metrics that directly compare generated and reference speech to produce a quality score; and segmentation issues, particularly in simultaneous S2ST. Furthermore, direct S2ST models are classified based on their application into offline and simultaneous models, which are discussed in detail in Section 6.2.1 and Section 6.2.2, respectively.

6.2. Application-based

Another way to categorize direct S2ST models is based on their mode of application: offline or simultaneous. The following subsections explain these models in detail. LLM-based S2ST can operate in either offline or simultaneous modes; however, we have covered it under the size-based characterization in Section 6.3. Offline models, as the name suggests, start decoding after having seen the entire utterance, whereas simultaneous (aka streaming models) models can start decoding only with partial utterances. As such, Simul-S2ST is amenable to real-time translation, dubbing, etc. In the following subsections, we will delve deeper into these two category models. In particular, we will discuss the architectural level changes made in vanilla Seq2Seq to devise the offline/Simul-S2ST models.

6.2.1. Offline S2ST models

As mentioned previously, offline direct S2ST models start decoding after having seen the entire utterance. The typical architecture of such models is shown in Fig. 4, which is based on Seq2Seq with attention (not shown in Figure). These models have *translation model* components besides speech encoder and vocoder (for speech synthesis). Dashed arrow components are optional and used in a few existing works. This

section will discuss various design choices used by existing works for translation model components as well as vocoder. Style adapters and duration predictors will also be explained as they play an important role in designing offline direct S2ST models.

Encoder Architecture: Some models use separate *speech encoders* to convert raw speech into spectrograms or discrete units, which are then fed into the translation model's encoder (Jia et al., 2019b; Diwan et al., 2024). In other cases, when the speech encoder is integrated with the translation module's encoder, raw speech is fed directly into the encoder of the translation module (Popuri et al., 2022; Li et al., 2023). For example, Translatotron 1 (Jia et al., 2019b) uses a spectrogram as input fed to the LSTM encoder. Conformer (Gulati et al., 2020) is used as the encoder in Translatotron 2 and 3 (Jia et al., 2022b; Nachmani et al., 2024). Essentially, recent direct S2ST models rely on transformers and their variants for encoder (Lee et al., 2022b; Li et al., 2023). To handle large speech inputs, there is often a downsampling module pre-pended before the encoder. 1-D CNN, Connectionist Temporal Classification (CTC) (Graves et al., 2006), and adapters are commonly used for length mismatch (Lee et al., 2022b; Popuri et al., 2022). As shown in Fig. 4, there is *optional* auxiliary task module attached to the encoder. The auxiliary task again relies on transformer blocks to predict source and/or target phonemes/char/word or discrete units depending upon the text data availability (Jia et al., 2022b; Lee et al., 2022b). Dual-encoders disentangling semantic and acoustic information have been proposed (Le et al., 2024).

Decoder Architecture: Similar to the encoder, the decoder of the translation model outputs either spectrogram (Jia et al., 2019b) or discrete unit (Lee et al., 2022b; Duquenne et al., 2023; Li et al., 2023). Decoders in the existing works have been designed either in a **Autoregressive** (AR) or **Non-autoregressive** (NAR) way, as explained below. Further, they could be **single-pass** or **double-pass** decoders. Double-pass decoders employ a linguistic and an acoustic decoder (Jia et al., 2022b; Inaguma et al., 2023; Li et al., 2023), contrary to single-pass decoders with only a semantic decoder. Due to multi-level speech processing and the ability to solve complex *multimodal distribution*,[3] double-pass decoders are shown to outperform single-pass decoders (Jia et al., 2022b; Inaguma et al., 2023).

In AR decoding, output tokens are predicted in a *causal* manner, i.e., current output depends only on the previously generated tokens. LSTMs and transformer-based decoders are preferred choices for autoregressive decoding. For example, unit mBART is used as a unit decoder in Popuri et al. (2022), Diwan et al. (2024) whereas Translatotron 1 and 2 use LSTMs. The CTC-based decoder is optionally added with the decoder as an auxiliary task (Lee et al., 2022b). Despite high-quality translations, two-pass autoregressive decoders suffer from *high-latency* issue because of their sequential generation.

NAR S2ST models have been proposed to address the high-latency decoding problem. NAR models can generate target speech in parallel by assuming conditional independence among the outputs. However, it becomes challenging to address the multimodal problem of the target speech via NAR compared to AR (Gu et al., 2018). To solve the linguistic modality problem, directed acyclic transformer (Huang et al., 2022) and FastSpeech 2 as two-pass decoders are used by DASpeech (Fang et al., 2023) whereas knowledge-distillation from AR model to NAR model is used in Huang et al. (2023c).

Besides AR and NAR decoding, there are other possible decoding choices. For example, it has been proven that translating multiple sentences of varying lengths can improve performance (Huang et al., 2023c). As such, **Top-K** length beam candidates are selected and decoded in parallel. **Noisy parallel decoding** is another option that uses AR decoding as a teacher to capture the more accurate optimum of the target distribution and compute the best translation for each fertility sequence (Huang et al., 2023c).

Vocoder: Following the translation module, a vocoder is responsible for synthesizing speech from the decoder's output, which can be in the form of text, discrete units, or spectrograms. Recent advancements in speech synthesis have led to the adoption of state-of-the-art vocoders. For instance, a majority of studies, including (Lee et al., 2022b; Popuri et al., 2022; Dong et al., 2022; Diwan et al., 2024), utilize HiFi-GAN (Kong et al., 2020) as the vocoder of choice. Other speech synthesis models include non-autoregressive Transformer stack initialized by VQ-VAE (Li et al., 2023), Light Convolution Blocks (Zhang et al., 2022), Non-Attentive Tacotron (NAT) (Shen et al., 2021) which has duration prediction module, an LSTM module, and a residual convolutional block (Jia et al., 2022b). Traditional estimation algorithms such as Griffin Lim (Griffin and Lim, 1984) may also be used to convert the mel-spectrogram to waveform. To support multilingual S2ST, multilingual vocoders have been trained with language embedding and language ID as an auxiliary loss (Gong et al., 2023).

Style Adapter: One of the desired characteristics of S2ST systems is that the translated speech should preserve (if need be) the original speaker's paralinguistic features, such as speaking rate, tone, intonation, etc. To this end, existing works insert *style adapter* layer between the translator's encoder and decoder. *Speech normalization* (Lee et al., 2022b) is one such technique to adapt to the source speaker's voice. The key idea is to extract discrete units from the reference speaker and then train the speech normalization module (which consists of pre-trained HuBERT + CTC) using multi-speaker data as input and reference discrete units as targets. For example, Translatotron 1 uses a discriminatively pre-trained encoder on speaker verification to condition the decoder for voice preservation. Due to **voice cloning** issue, the aforementioned technique is unsuitable for production deployment. An approach to address this issue is to train on the same speaker's speech on the source and target side so that the model can transfer prosodic information at inference time (Jia et al., 2022b). A side benefit that comes for free is that the S2ST model does not require speaker segmentation for translating multi-speaker utterances. Interestingly, recent S2ST models implicitly model the para-/non-linguistic characteristics during decoding (Dong et al., 2024; Peng et al., 2024a).

Duration Predictor: Duration predictor is a module often applied (but not always) along with the speech synthesizer. Its job is to predict the duration of each output element– be it a phoneme or discrete unit, which is later converted to a waveform. For example, Translatotron 2 uses NAT TTS (Shen et al., 2021), which predicts the duration of each element followed by upsampling. However, unlike NAT, it optimizes $\mathcal{L}^2$ loss on the total duration predicted instead of *per-phoneme* level which is costly to obtain. TTS models like Fastspeech 2 (Ren et al., 2020) have an in-built duration predictor, which has been used by Lee et al. (2022a). Works employing unit-to-speech conversion (Lee et al., 2022b; Inaguma et al., 2023; Fang et al., 2023; Zhu et al., 2023; Shi et al., 2023) via HiFi-GAN TTS enhance the latter via duration prediction module from (Ren et al., 2020).

6.2.2. Simultaneous (simul-) S2ST models

The models following the offline setting consider all the input speech to be available before commencing the translation process, as discussed in Section 6.2.1. In contrast, simultaneous models initiate translation upon receiving partial input speech (Agarwal et al., 2023). Let us assume the duration of the source and target speech are $\mathcal{T}_{src}$ and $\mathcal{T}_{tgt}$, respectively. The input begins at $t = 0$, and the translation output starts at $t = \mathcal{T}_0$. In the case of the offline model, it has $\mathcal{T}_0 \geq \mathcal{T}_{src}$, whereas, for the simultaneous model, it has $\mathcal{T}_0 < \mathcal{T}_{src}$ (Ma et al., 2021). Simul-S2ST models aim to deliver accurate speech translation with minimal delay, enabling seamless communication. However, achieving

[3] S2ST is considered a multimodal problem due to (a) linguistic diversity during translation, and (b) diverse acoustic conditions, e.g., pitch, duration, energy, etc.

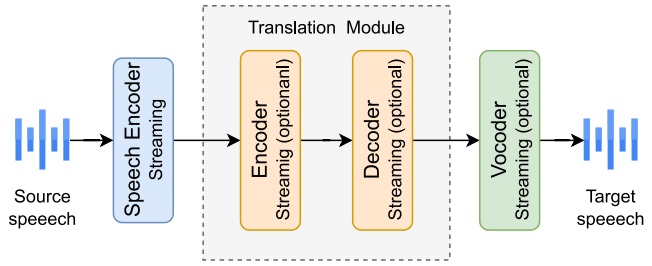


Fig. 5. Typical architecture of simultaneous direct S2ST model.

high translation quality while reducing latency remains a formidable challenge, making Simul-S2ST a critical problem in the field (Dugan et al., 2023; Ma et al., 2021). The Simul-S2ST problem faces several issues in practical implementation, including reordering, acoustic ambiguity, variable speech rates, and handling long input sequences. One of the most important decisions in Simul-S2ST is to decide when to start the translation while balancing latency and quality. A Simul-S2ST model typically consists of a *speech encoder*, a *translation module*, and a *vocoder*, as depicted in Fig. 5.

Streaming Encoder: The speech encoder is similar to the offline model's encoder shown in Fig. 4 and described in Section 6.2.1, with the addition of streaming capabilities, distinguishing it from the offline version. To make the model simultaneous, the speech encoder must have a streaming capability, while other modules may not. A fundamental approach to making the S2ST model simultaneous is to make the modules causal. However, while fully causal modules can achieve minimal delay, they can compromise translation accuracy (Rybakov et al., 2023a). An alternative approach is to use the simultaneous policy (SiP), such as the wait-k strategy in the speech encoder, where the model waits for k frames before processing and passing the data to the translation model (Ma et al., 2021; Dugan et al., 2023; Xu et al., 2024). Speech encoders, such as the mel-frontend used in Rybakov et al. (2023a), Agranovich et al. (2024), can be employed in a real-time manner when other modules can balance the latency-quality tradeoff using suitable policy effectively. The models in Dugan et al. (2023), Xu et al. (2024) use only a simultaneous speech encoder, while other modules are directly adopted from offline models.

The encoder of the translation module receives input from the speech encoder. Conformer (Gulati et al., 2020) and Transformer (Vaswani et al., 2017) architectures are commonly preferred. Some models make the encoder causal to enable streaming capabilities, restricting it to attend only to the current and past input sequences (Barrault et al., 2023a; Rybakov et al., 2023a). Encoders may also incorporate wait-k SiP, similar to speech encoders, when the speech encoder operates in real-time. To further optimize the latency-quality tradeoff, SiP methods such as Lookahead Self-Attention (LAS) and Lookahead Self-Attention with a Lookahead Stacker (LAS_LS) (Rybakov et al., 2023a) are employed, enabling the encoder to utilize a limited number of future sequences. The SiP V-MMA (Variational Monotonic Multihead Attention (Ma et al., 2021)) considers every attention head as independent monotonic attention and models the alignment between the source and target sequences using latent variables instead of recurrent estimation, which leads to an inferior estimate of the alignment matrix.

Streaming Decoder: The decoders of Simul-S2ST systems can utilize offline architectures, as discussed in Section 6.2.1, real-time streaming approaches (Rybakov et al., 2023a), or specific SiPs to achieve better latency-quality trade-offs (Barrault et al., 2023a). The SiPs employed in the encoder, such as wait-k, LAS, or LAS_LS, are directly adapted for use in the decoder to enable streaming capabilities. Additionally, Efficient Monotonic Multihead Attention (EMMA) (Ma et al., 2023) is incorporated in the decoder to reduce latency and preserve the paralinguistic features of speech (Barrault et al., 2023a). In Agranovich et al. (2024), the decoder employs *wait-k attention*, which processes speech by accumulating k encoded speech frames at a time. To provide a contextual reference, (Biadsy et al., 2019; Agranovich et al., 2024) utilize PreNet and PostNet with residual connections in the decoder. PreNet and PostNet are lightweight neural network modules comprising a few layers. Furthermore, several studies have implemented concurrent execution of different modules to minimize waiting time, effectively reducing overall latency (Biadsy et al., 2019; Dugan et al., 2023; Agranovich et al., 2024). Traditional Simul-S2ST models (cascaded Simul-S2ST) have studied latency-quality trade-off under various policies (Zheng et al., 2020; Dugan et al., 2023) and find that no-single best-policy exist for all languages. Hence, it is recommended to tune the policy per language basis.

Streaming Vocoder: Simul-S2ST optionally integrates vocoders from offline models, but significant efforts have been made to develop streaming vocoders to optimize the tradeoff between latency and quality. Vocoders can be broadly categorized into two types: traditional iterative phase reconstruction methods, such as Griffin-Lim (GL), which use standard signal processing algorithms and do not require prior knowledge of target signals, and DL-based models like MelGAN and HiFiGAN, which rely on target signals during training (Masuyama et al., 2021). Real-time streaming vocoders often suffer from lower perceptual quality in synthesized speech (Rybakov et al., 2023b). However, introducing a *one-hop lookahead* (12.5 ms) has significantly enhanced speech quality. While GL underperforms compared to DL-based models in terms of quality, it is favored for on-device applications due to its low memory and computational requirements (Rybakov et al., 2023b). Streaming MelGAN, known for its parallel processing capabilities, has been employed in Rybakov et al. (2023a), Agranovich et al. (2024) to further improve quality. Additionally, these studies used a two-hop lookahead instead of a single-hop, achieving better-synthesized speech quality. In Ma et al. (2020a), the HiFiGAN unit vocoder (non-autoregressive) is utilized in its offline version, specifically trained on shorter-duration audio clips.

6.3. Size-based

Another way to characterize S2ST models is based on their size: Small Language Models (SLMs) and Large Language Models (LLMs). The primary criterion for this classification is the number of trainable parameters in the language model's neural network. Although there is no strict threshold, several articles consider models with fewer than a few billion parameters as SLMs, while LLMs typically range from tens of billions to trillions of parameters (Shone, 2024; instinctools, 2024; Greer, 2024). SLMs and LLMs each have their own advantages and disadvantages, and their suitability varies depending on the application (Greer, 2024). Direct S2ST models based on SLMs and LLMs are described in detail in the following subsections.

6.3.1. Small language model (SLM)-S2ST

A language model is commonly considered an SLM when its number of parameters is less than a few billion. In the present work, we classify a direct S2ST model as an SLM if it has around 1 billion parameters unless explicitly specified as an LLM. SLMs are considered a good option when constraints include limited computational resources, lower latency requirements, limited training data, the need for task-[redacted] and reduced training time (Greer, 2024). All SLM-based direct S2ST models are included under the application-based categorization: offline and simultaneous—as discussed in detail in Section 6.2.

6.3.2. Large language models (LLM)-S2ST

LLMs have a significantly higher number of trainable parameters than SLMs, typically ranging from a few billion to several trillion. These

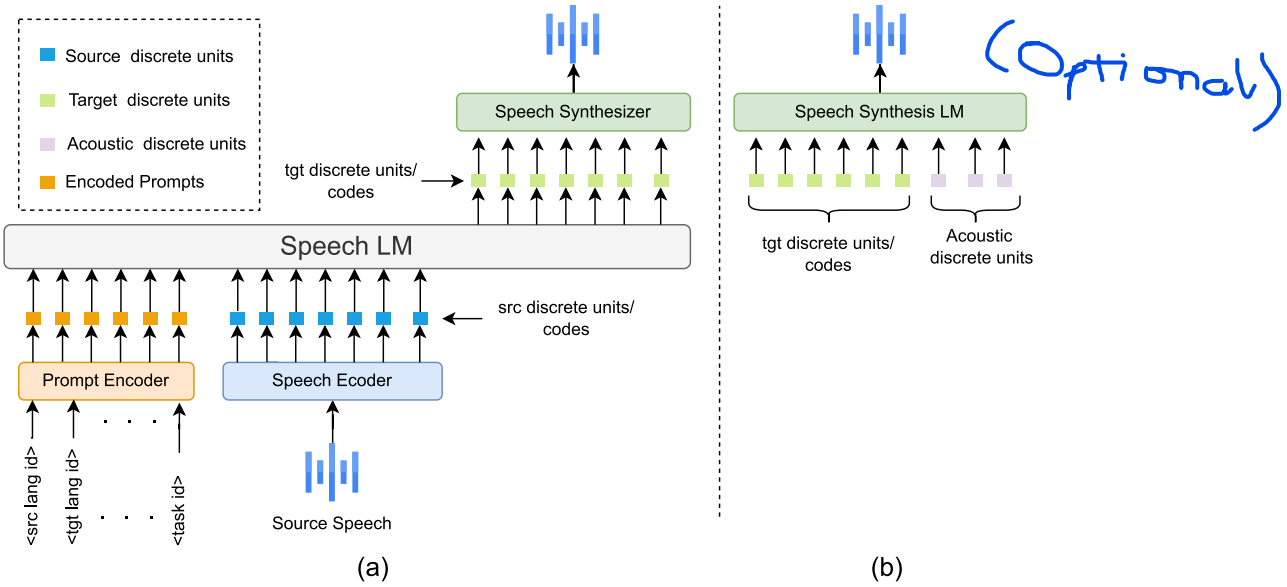


Fig. 6. Typical architecture of LLM-based direct S2ST: (a) Complete architecture incorporating a speech synthesizer without voice/style preservation, and (b) A speech synthesizer that utilizes source acoustic units to ensure voice/style preservation.

models require vast amounts of data and substantial computational resources. However, they are well-suited for complex NLP tasks and have diverse applications. A single model can perform multiple tasks through prompting and can adapt to new situations using fine-tuning with limited examples.

Recent advancements in speech foundation models, such as Whisper (Radford et al., 2023), have achieved remarkable performance by training on approximately 680K hours of weakly supervised and diverse speech data. The Whisper model delivers competitive results compared to state-of-the-art ASR systems and demonstrates superior robustness—without relying on SSL or self-training. Its strength lies in joint multilingual and multitask training, significantly enhancing its capabilities of zero-shot ST. Whisper uses a standard Transformer architecture, and while its model weights are open-sourced, the training code remains proprietary. An Open Whisper-style Speech Model (OWSM) (Peng et al., 2023, 2024b) is an open-source alternative to Whisper model, trained on publicly available data and released in multiple parameter sizes. It aims to serve as a fully open-source speech foundation model by providing both model weights and training code. While the initial version of OWSM used a Transformer encoder, the more recent OWSM v3.1 (Peng et al., 2024c) adopts E-Branchformer (Kim et al., 2023), an enhanced variant of Branchformer (Peng et al., 2022). These foundation models pave the way for developing zero-shot or few-shot S2ST systems by leveraging large-scale speech LLMs.

The recent success of Generative Pre-Trained Transformers (GPT) (Radford et al., 2018; Brown et al., 2020; Ouyang et al., 2022) and BERT models (Devlin et al., 2019) over various NLP tasks gives rise to what we know as LLMs. These models exhibit in-context learning (ICL) when trained on vast datasets (Dong et al., 2024; Peng et al., 2024a). Extensive training unlocks their latent *emergent abilities* (Wei et al., 2022), enabling them to perform few-shot and zero-shot learning through prompting. It alludes to using LLMs for S2ST tasks as well.

An LLM-based direct S2ST model typically follows the architecture shown in Fig. 6(a). Instead of a traditional encoder–decoder setup, it uses a decoder-only speech LM, which is prompted to generate speech in the target language. However, a key challenge arises because LLMs process discrete tokens, whereas speech consists of continuous values that these models cannot directly utilize. To address this, several studies (Wu et al., 2023; Zhang et al., 2023; Dong et al., 2024) employ discrete-unit representations of source and target speech. The speech encoder in Fig. 6(a) converts continuous speech into discrete tokens using SSL models such as HuBERT (Hsu et al., 2021), Wav2vec2.0 (Baevski et al., 2020), and WavLM (Chen et al., 2022), in conjunction with k-means clustering. The Speech LM then performs unit-to-unit translation (U2UT), transforming the source discrete units into target discrete units. The speech synthesizer subsequently converts these translated tokens into target speech. Optionally, the synthesizer module can incorporate acoustic units from the source speech for voice preservation or style transfer, as shown in Fig. 6(b) (Peng et al., 2024a; Dong et al., 2024).

The effectiveness of LLM-based S2ST lies in several strategies, such as (a) what prompt to use? (b) how to do prompt-tuning? (c) which LM to use? Expressive S2ST (Gong and Veluri, 2024) claims to preserve the speaker style without relying on aligned speech-text data or speaker-aligned speech. There are still some questions that need to be answered. For example, what is the best strategy for prompt design, and how can they be pre-train/fine-tuned parameter-efficiently for S2ST tasks? Further, the use of LLMs for Simul-ST has been recently proposed (Ouyang et al., 2024; Cheng et al., 2024) and it remains to see how to adapt Simul-MT to Simul-S2ST. In the following subsections, prompting techniques and LM architectures are explained in detail.

Prompting Techniques: Prompting is a technique used to supply additional information, typically in the form of tokens alongside the input, to guide the model towards generating the desired output (Lester et al., 2021). Prompts help leverage the inherent capabilities of models, such as multimodal capability (Dong et al., 2024), multilingual understanding (Zhang et al., 2023), and voice preservation (Peng et al., 2024a).

Two types of prompts are commonly used: *hard prompts* and *soft prompts*. In a hard prompt, relevant parameters (such as language or task IDs) are directly provided as discrete tokens. In contrast, a soft prompt consists of trainable embeddings prepended to the input and learned during training. For example, in a multilingual model, a hard prompt may look like:

Prompt1 : ⟨⟨translate⟩, [src_LID], src_unit, [tgt_LID], tgt_units⟩

Here, language IDs (LIDs) are explicitly provided. In contrast, for voice preservation, acoustic units may be used as prompts, consisting of embeddings similar to semantic units (Dong et al., 2024).

Prompt2 : ⟨src_semantic_unit, tgt_semantic_unit, src_acoustic_unit⟩

LLM-based models use two types of speech units extracted from speech representation models: *semantic units* and *acoustic units*. Semantic units are obtained from models such as HuBERT (Hsu et al., 2021) and w2v-BERT (Chung et al., 2021), while acoustic units are extracted from models like SoundStream (Zeghidour et al., 2021) and EnCodec (Défossez et al., 2022). Semantic units are responsible for preserving linguistic content during translation, while acoustic units capture aspects of voice preservation, including speaking style, emotion, rhythm, and dialect. Source or target acoustic units are included in the prompt, depending on whether to preserve speech according to source or target speech. The target speech acoustic units are provided as a prompt if output speech needs the voice of the target speech. However, mostly in the case of acoustically non-aligned data, the source speech acoustic units are used as a prompt to transfer the natural speech qualities from source to target speech (Dong et al., 2024).

Multi-step prompting and *Chain-of-Thought (CoT)* prompting are commonly employed for voice preservation. In multi-step prompting, a single task is broken down into multiple sub-tasks, each handled through separate prompts to achieve the desired output. For each sub-task, a separate language model or a distinct execution stage (training or inference) may be required. For example, $S_{\text{src}} \rightarrow S_{\text{tgt}}$ represents the translation of source semantic units to target semantic units, and $S_{\text{tgt}} + A_{\text{prompt}} \rightarrow A_{\text{tgt}}$ transforms the target semantic units into target acoustic units (Polyak and Wolf, 2019; Zhang et al., 2023). In CoT prompting, a step-by-step reasoning process is followed, where the output of each step informs the next. All steps are represented within a single prompt sequence and are trained end-to-end (Gong and Veluri, 2024).

LM Architectures: The LLM-based S2ST models commonly use Transformer-based decoder-only LM. However, multiple LM is used separately for semantic translation, acoustic translation, and duration prediction (Dong et al., 2024). The model proposed by Peng et al. (2024a) uses a single LM for both semantic-to-semantic and semantic-to-acoustic translation. This LM operates in an AR manner during the first step to generate the initial stream of 8-stream acoustic units and then switches to an NAR mode in the second step to generate the remaining acoustic streams. The first and second steps of AR and NAR share the same model parameters but require separate training stages. The model proposed in Gong and Veluri (2024) also uses a single language model internally divided into two components: an AR decoder layer and a NAR decoder layer. Similar to Peng et al. (2024a), it operates in two steps; instead of separate training for each step, it trains the LM E2E using CoT prompting. The latter model, which employs E2E training using CoT prompting with a single LM, demonstrates better performance than others, as evidenced by the ablation study in Gong and Veluri (2024). A denoising autoencoder, mBART, has also been used as the backbone LM in SpeechGen (Wu et al., 2023).

7. Training strategies

The training of direct S2ST typically follows the process used for DL models (Bengio, 2012). Along with model architecture and optimization algorithms, a key factor influencing the model's performance is handling input/output speech, referred to as I/O representation. There are two primary approaches for I/O representation of speech: (1) *raw waveform or spectrogram*, and (2) *discrete units or tokens*.

When using a spectrogram as the I/O representation, the speech frames x, represented as a spectrogram, are processed by encoder–decoder models as follows: The input speech signal x is converted into a sequence of embeddings: $\mathcal{H} = \text{Encoder}(x) = \{h_1, h_2, \ldots, h_T\}$ where $h_i \in \mathbb{R}^d$ is a hidden embedding vector, and T represents the total number of hidden vectors. The number of input speech frames (k) and the number of hidden vectors (T) may differ, as the encoder often includes convolutional layers that reduce the length of the input sequence. The decoder then generates the target spectrogram from the hidden state vectors as follows: $\hat{y} = \text{Decoder}(\mathcal{H})$ where $\hat{\mathbf{y}}$ represents the predicted spectrogram. The loss is computed between the input spectrogram x and the predicted spectrogram y to optimize the likelihood, as shown in (2). Commonly used loss functions include Mean Squared Error (MSE) and Mel-Cepstral Distortion (MCD) (Jia et al., 2019b).

In the case of discrete unit representation, speech is converted into discrete units using speech representation models like HuBERT, as described in Section 5.2. There are two primary approaches to utilizing discretized speech: (1) using only the target as discrete units and (2) using both the source and target as discrete units. In the first approach, the source speech, represented as a spectrogram, is translated into target speech units. This method is commonly known as Speech-to-Unit Translation (S2UT) (Lee et al., 2022a; Popuri et al., 2022). The second approach converts both the source and target speech into discrete units. The source units are then translated into target units, called Unit-to-Unit Translation (U2UT) (Gong et al., 2023; Kaur et al., 2024; Diwan et al., 2024). In both approaches, the target consists of discrete tokens, making cross-entropy loss the preferred choice for the loss function (Lee et al., 2022a; Popuri et al., 2022).

Apart from the utilization of parallel speech as techniques mentioned above and representation learning as mentioned in Section 5.2, it is also essential to how supervised text and unsupervised speech and text can be utilized through various techniques such as Pre-training, self-supervised, unsupervised, and weakly supervised training primarily used to solve data scarcity as mentioned in Section 3. Therefore, we split our discussion on training based on the availability of *external data*.

7.1. Training with external data

Training of direct S2ST models optimizes the negative conditional log-likelihood on the supervised speech data as given in (2). However, there are cases when external resources such as source and/or target transcripts are available in addition to target speech. Therefore, a natural question arises on how to leverage such external data. An approach, *Multitask Learning* (MTL), where different sub-tasks are optimized simultaneously employing the E2E training approach (Tampuu et al., 2020; Prabhavalkar et al., 2024) are employed to utilize the supervised transcripts in case of high-resource written languages.

As shown in Fig. 4, the architecture incorporates auxiliary decoders to leverage available labeled transcripts for supervising the model. In Jia et al. (2019b), source and target auxiliary decoders are applied to different layers of a stacked LSTM encoder. The assumption was that the initial layers are more representative of the source language, while the deeper layers better capture the target language. In Kano et al. (2021), auxiliary decoders have been employed between the encoder and decoder by introducing an attention layer and connecting them using Transcoders. In addition to the primary loss ($\mathcal{L}_{S2S}$) defined in (2), auxiliary decoding losses $\mathcal{L}^s_{rec}$ and $\mathcal{L}^t_{rec}$ similar to the (5), computed for the recognition of source and target text, respectively, are also included. The overall loss is then jointly optimized as: $\mathcal{L}_{total} = \mathcal{L}_{S2S} + \mathcal{L}^s_{rec} + \mathcal{L}^t_{rec}$. Incorporating textual supervision through MTL leads to an enhancement in the translation quality of the models (Jia et al., 2019b, 2022b; Lee et al., 2022a; Popuri et al., 2022; Shi et al., 2023).

There are several issues present in Translatotron (Jia et al., 2019b) such as: (1) auxiliary tasks for textual supervision are underutilized, as the learnings from auxiliary attention are not fully transferred to the main module, (2) the model faces difficulty mapping lengthy source speech spectrograms to target spectrograms, (3) over generation and under generation problem due to attention mechanism (Ren et al., 2019; Zheng et al., 2019; Shen et al., 2020). The existing bottlenecks are mitigated through architectural changes in Translatotron2 (Jia et al., 2022b) with four sub-modules. The single attention mechanism, based on Transformer (Vaswani et al., 2017) alleviates the issue of lengthy spectrogram sequences by aligning them with shorter-length phonemes. Additionally, a Non-Attentive Tacotron (NAT) (Shen et al., 2021) based TTS is employed to mitigate problems of over-generation

and under-generation. Several models following the above architecture also use unsupervised data using techniques described in Section 3 to improve the performance of models (Dong et al., 2022; Jia et al., 2022a; Nachmani et al., 2024).

7.2. Training w/o external data aka textless training

Compared to high-resource written languages, low-resource unwritten languages lack transcripts, making traditional approaches challenging. Recent efforts in direct S2ST modeling have proposed *textless* training for these languages. One approach involves training the model directly on speech spectrograms; however, this method often struggles to learn generalized patterns without textual supervision. Alternatively, textless training can leverage SSL models (e.g., HuBERT Hsu et al., 2021, Wav2vec2.0 Baevski et al., 2020, WavLM Chen et al., 2022, XLS-R Babu et al., 2022) or unsupervised models (e.g., VQ-VAE Oord et al., 2017) for speech tokenization, as discussed earlier in the section. After converting continuous speech into discrete tokens (similar to text), enabling the application of textual NLP tools to speech (Lee et al., 2022b).

For example, The textless model in Zhang et al. (2021) comprises three modules: converter, translator, and inverter. The converter transforms target speech into discrete units, the translator translates source speech into target discrete units, and the inverter inverts the predicted discrete units back into a speech waveform. Target discrete units enable the model to use the cross-entropy loss for model optimization. This architecture is beneficial for untranscribed languages (Tjandra et al., 2019; Zhang et al., 2021; Lee et al., 2022b; Huang et al., 2023c) or speech datasets without labeled transcripts (Lee et al., 2022a; Huang et al., 2023c). The process of acquiring phonetic knowledge from languages that share syntactic similarities and have a written form might aid in learning representations for unwritten languages (Yallop and Fletcher, 2007; Kuhl et al., 2008). Nevertheless, the extent to which this assistance proves effective depends on the degree of similarity between the languages. Hence, leveraging the benefits of related languages with writing systems, XL-VAE (Zhang et al., 2021) is trained in a supervised manner by using phonemes from the related language as targets.

Some works (Huang et al., 2023a) augment S2UT with visual data and apply SSL pre-training with joint speech-visual data and cross-modal knowledge-distillation for textless training. U2UT with BT (Diwan et al., 2024) and diffusion generative models (Zhu et al., 2023) are some other recent efforts towards textless training. U2UT model in Kim et al. (2024a) used textless training for many-to-many S2ST. Despite their less reliance on parallel speech corpora, these models still underperform compared to models trained on a vast amount of external text data. Therefore, there is room for improvement by training better, larger unit-LMs and over multiple languages for semantic and acoustic property transferability across languages.

8. Experimental results and discussion

This section discusses the performance of the models, which are evaluated using various performance parameters. Offline models are typically evaluated based on translation accuracy and naturalness. Most models use ASR-BLEU, as described in Section 4.1, to evaluate translation accuracy and MOS to measure naturalness; however, MOS scores have been reported in only a limited number of studies. More recent models have begun reporting the BLASER score for translation accuracy, but this metric is not included in Table 2 because the available results are limited and not directly comparable. The open-source S2ST datasets listed in Table 1 and several proprietary datasets not included in the table, such as Conversational (Jia et al., 2019a) and Fisher (Cieri et al., 2004), are utilized for model training and performance evaluation.

Table 2 compares the performance of cascade and direct S2ST offline models from various studies. Results show a significant performance gap in ASR-BLEU scores between textless direct models without external data and cascade models (ID:1-4). The discrete unit-based Translatotron 2 (Li et al., 2023) (textless setup with external data), which utilizes the pre-trained w2v-BERT (Chung et al., 2021) as its encoder, outperforms other direct and cascade models in the Fisher Es→En group (ID:5-10), which demonstrates the effectiveness of speech representation learning models for S2ST. In Inaguma et al. (2023), Zhang et al. (2024), with two-pass decoders, surpasses all counterparts in its datasets and language directions and demonstrates performance comparable to cascade models (ID:14, 18, 22, 28, 38). Utilizing text data through MTL and external data via pre-training benefits models (Jia et al., 2019b; Popuri et al., 2022; Lee et al., 2022a). Models with two-pass decoders outperform single-decoder models (Jia et al., 2022b; Inaguma et al., 2023; Li et al., 2023). However, the model proposed in Fang et al. (2024b), while claiming to be a direct model and to outperform two-pass decoder-based direct models (Jia et al., 2022b; Inaguma et al., 2023), is in fact closer to a cascade (ST+TTS) architecture rather than a true direct S2ST model. For this reason, its results are not included in Table 2. The MOS of direct models is slightly lower than that of cascade models, perhaps due to training with synthesized target speech (ID:5-7,11,12), which limits their scores. MOS also depends on vocoder performance; lower translation quality negatively impacts MOS ratings (Jia et al., 2019b). Using natural speech in the target can surpass the MOS of cascade models (ID:18, 24).

Table 3 summarizes results for Simultaneous and LLM-based models. To effectively compare the performance of Simul-S2ST models, several standard criteria must be established, including the dataset, language pair and direction, SiP (Simultaneous Policy), and hyperparameters such as the value of k in wait-k, as well as latency metrics. Similarly, comparison of LLM-based models is feasible only when they are evaluated on common language pairs, directions, and datasets. However, due to the lack of consistent results, drawing definitive conclusions is not currently possible.

A standard dataset and language pair are necessary to compare the models fairly. Therefore, we implemented some existing models on the Es→En language pair of the CVSS-C dataset, as shown in Table 4, acknowledging that only some models are implemented due to code reproducibility issues and limited computing resources. The Fairseq library (Ott et al., 2019) is used on a single NVIDIA Quadro GV100 GPU machine. The models (ID: 53-58) are trained from scratch with the provided hyperparameter settings. The results indicate that the performance of the direct model is deficient without MTL (ID: 55). However when textual supervision is introduced through MTL, the performance of the discrete unit-based model (Lee et al., 2022a) shows significant improvement (ID: 46), while the spectrogram-based model (Jia et al., 2019b) still struggles with the provided amount of data (ID: 56). The model by Popuri et al. (2022), which uses a pre-trained encoder and decoder and fine-tunes on CVSS-C dataset, outperforms all models trained without pre-training (ID: 59-62) and also outperforms the cascade 2-stage models, but falls short compared to the cascade 3-stage model.

9. Application issues

S2ST has many application domains, such as movie dubbing and real-time translation. However, adapting to these applications' specific conditions and requirements presents several challenges. For instance, in movie dubbing, tasks such as handling *cross-talk or overlapping speech*, extracting clean voices in noisy environments, and removing background music must be performed accurately by the S2ST models. Translating dialogue while preserving each character's unique voice and local accent is also a significant challenge. *Speaker diarization* is also essential for accurately identifying each speaker (Tranter and Reynolds, 2006). Moreover, translation must be adapted to the domain

Table 2
Performance comparison of offline direct and cascade S2ST models. The columns under "Direct"provide details about the direct models, while the columns under "Cascade"present the performance of the cascade baseline models. Only the direct S2ST models with at least one comparable result based on the dataset and language direction are included. Models that do not have results on the standard dataset are excluded. The BLEU scores indicate translation quality, MOS scores assess the subjective speech quality, and the "Textless"column shows whether text data was used during training. There are two types of highlighted values: bold and bold with underlines. A bold value indicates that the model is outperforming other models within the group of direct models. A bold, underlined value indicates the winner among direct and cascade within the group (Kim et al., 2024c).

					Direct		Cascade Baseline		
ID	Paper	Dataset	Lang.	Textless	BLEU	MOS	BLEU	MOS	Model
Models without external data									
1	Jia et al. (2019b)	Fisher	Es→En	✓	0.6	✗	**41.4**	4.09	ST+TTS
2	Zhang et al. (2021)	Fisher	Es→En	✓	**9.35**	✗	✗	✗	✗
3	Huang et al. (2023c)	CVSS	En→Es	✓	12.77	✗	**32.86**	✗	ST+TTS
4	Kim et al. (2024c)	CVSS	En→Es	✓	**18.72**	✗	✗	✗	✗
Models with external data									
5	Jia et al. (2019b)	Fisher	Es→En	✗	30.1	3.69	41	4.09	ST+TTS
6	Jia et al. (2022b)	Fisher	Es→En	✗	42.9	3.98	43.3	4.04	ST+TTS
7	Lee et al. (2022a)	Fisher	Es→En	✗	39.9	3.41	43.9	3.43	ASR+MT+TTS
8	Dong et al. (2022)	Fisher	Es→En	✗	43.6	3.59	41.4	✗	ST+TTS
9	Shi et al. (2023)	Fisher	Es→En	✗	41.5	✗	✗	✗	✗
10	Li et al. (2023)	Fisher	Es→En	✓	**50.3**	✗	✗	✗	✗
11	Jia et al. (2019b)	Conversational	Es→En	✗	42.7	4.08	48.7	4.32	ST+TTS
12	Jia et al. (2022b)	Conversational	Es→En	✗	**55.6**	4.2	**58.8**	4.31	ST+TTS
13	Lee et al. (2022b)	CoVoST	Es→En	✓	16.3	✗	14.8	✗	ST+TTS
14	Inaguma et al. (2023)	CoVoST	Es→En	✗	**36.4**	✗	32.9	✗	ASR+MT+TTS
15	Popuri et al. (2022)	CoVoST	Es→En	✗	33.5	3.15	33.8	3.53	ASR+MT+TTS
16	Wang et al. (2024)	CVSS	Es→En	✗	23.41	3.87	✗	✗	✗
17	Nachmani et al. (2024)	CVSS	Es→En	✗	14.25		6.13(unsu.)	3.14	ASR+MT+TTS
18	Zhang et al. (2024)	CVSS	Es→En	✗	**27.25**	✗	✗	✗	✗
19	Lee et al. (2022b)	EuroPerl-ST	Es→En	✓	18.9	3.26	19.2	3.23	ST+TTS
20	Popuri et al. (2022)	EuroPerl-ST	Es→En	✗	28.6	3.15	29.1	3.53	ASR+MT+TTS
21	Wei et al. (2023)	EuroPerl-ST	Es→En	✗	24.6	✗	✗	✗	✗
22	Inaguma et al. (2023)	EuroPerl-ST	Es→En	✗	**33.1**	✗	**34.2**	✗	ASR+MT+TTS
23	Zhu et al. (2023)	EuroPerl-ST	Es→En	✓	14.8	✗	19.2	✗	ST+TT
24	Kim et al. (2024a)	EuroPerl-ST	Es→En	✓	21.3	✗	✗	✗	✗
25	Lee et al. (2022b)	EuroPerl-ST	En→Es	✓	22.7	4.11	21.7	4.12	ST+TTS
26	Popuri et al. (2022)	EuroPerl-ST	En→Es	✗	32.7	4.06	32.6	3.8	ST+TTS
27	Wei et al. (2023)	EuroPerl-ST	En→Es	✗	26.6	✗	✗	✗	✗
28	Inaguma et al. (2023)	EuroPerl-ST	En→Es	✗	**35.3**	✗	**36.8**	✗	ASR+MT+TTS
29	Zhu et al. (2023)	EuroPerl-ST	En→Es	✓	14.5	✗	21.7	✗	ST+TT
30	Kim et al. (2024a)	EuroPerl-ST	En→Es	✓	23.4	✗	✗	✗	✗
31	Lee et al. (2022b)	EuroPerl-ST	Fr→En	✓	**20.3**	3.27	19.8	3.22	ST+TTS
32	Zhu et al. (2023)	EuroPerl-ST	Fr→En	✓	15.2	✗	19.8	✗	ST+TT
33	Kim et al. (2024a)	EuroPerl-ST	Fr→En	✓	20.7	✗	✗	✗	✗
34	Lee et al. (2022b)	EuroPerl-ST	En→Fr	✓	18.7	2.87	18.5	2.44	ST+TTS
35	Zhu et al. (2023)	EuroPerl-ST	En→Fr	✓	13.6	✗	18.5	✗	ST+TT
36	Kim et al. (2024a)	EuroPerl-ST	En→Fr	✓	**21.2**	✗	✗	✗	✗
37	Popuri et al. (2022)	MUST-C	En→Es	✗	32.1	4.06	34.2	3.8	ASR+MT+TTS
38	Inaguma et al. (2023)	MUST-C	En→Es	✗	**36.4**	✗	30.8	✗	ASR+MT+TTS

Table 3
Performance description of direct Simul-S2ST models and LLM-based direct S2ST models. The table provides a detailed performance description and a comparison with the cascade model's performance. Some LLM-based models also show results while providing target oracle text as input to the model along with source speech. However, the table only shows results without Oracle target text as input to the model.

				Direct		Latency		
ID	Paper	Dataset	Lang. Pair	BLEU	MOS	Latency (ms)	Metrics	Remark
Simultaneous models								
39	Ma et al. (2021)	Fisher	Es→En	25.3/25.9	✗	3136/3215	AL	CA
40	Ma et al. (2021)	MuST-C	Es→En	11.5/12.2	✗	2706/2927	AL	CA
41	Agranovich et al. (2024)	MuST-C	En→Es	14.6/14.7	3.35	1100/2300	Total delay	k-125/k-150
42	Agranovich et al. (2024)	Conv.	Es→En	51.2	3.35	18	Total delay	Single Frame
43	Zhang et al. (2024)	CVSS	Es→En	20.6/21.68	✗	1522/2514	AL	Non-CA
44	Zhang et al. (2024)	CVSS	Fr→En	22.89/24.41	✗	1270/2326	AL	Non-CA
45	Zhang et al. (2024)	CVSS	De→En	14.56/15.83	✗	1688/2562	AL	Non-CA
LLM-based models								
46	Wu et al. (2023)	CoVoST	En→Es	15.9	✗	✗	✗	✗
47	Zhang et al. (2023)	EMIME	En→Zh	34.45	3.41	✗	✗	✗
48	Zhang et al. (2023)	EMIME	Zh→En	30.66	3.54	✗	✗	✗
49	Dong et al. (2024)	EMIME	Zh→En	29.40	4.10	✗	✗	✗
50	Peng et al. (2024a)	In-house	Es→En	24.78	✗	✗	✗	✗
51	Peng et al. (2024a)	In-house	En→Es	22.09	✗	✗	✗	✗
52	Gong and Veluri (2024)	In-house	Es→En	20.16	3.28	✗	✗	✗

Table 4
Results of Es→En language pair on CVSS-C dataset. (S2UT: Speech-to-unit Translation, MTL: Multitask Learning, TF: Transformer), LNA: LayerNorm Attention Module, LNA-E: LNA, and self-attention of the encoder and full decoder are fine-tuned. LNA-D: The whole encoder and the LNA and both encoder and self-attention in the decoder are fine-tuned. LNA-D,E: LNA parameters are fine-tuned on the encoder and decoder. Full: all the parameters of the encoder and decoder are fine-tuned.

ID	Models	Architecture	#Parameters	BLEU
53	ASR+MT+TTS (Vaswani et al., 2017)	Cascade	97M	**32.6**
54	ST+TTS (Wang et al., 2021b)	Cascade	42M	19.80
55	S2UT (Lee et al., 2022a) w/o MTL	Direct	42.8	0.86
56	S2UT (Lee et al., 2022a) w/ MTL	Direct	88.1M	**14.60**
57	TF-Translatoron (Jia et al., 2019b) w/o MTL	Direct	63.9M	0.40
58	TF-Translatoron (Jia et al., 2019b) w/ MTL	Direct	132.9M	3.78
59	S2UT(Wav2vec-L+mBART)(Popuri et al., 2022) LNA-D	Direct	827M	21.19
60	S2UT(Wav2vec-L+mBART)(Popuri et al., 2022) LNA-E	Direct	827M	20.09
61	S2UT(Wav2vec-L+mBART)(Popuri et al., 2022) LNA-D,E	Direct	827M	23.27
62	S2UT(Wav2vec-L+mBART)(Popuri et al., 2022) Full	Direct	827M	**23.47**

and audience, ensuring the semantic meaning remains consistent while remaining engaging and culturally appropriate for the target audience. Another major challenge is achieving *isochrony* between the translated speech, source speech, and visual content. Proper time alignment is difficult, as the same content may differ in duration in different languages. The IWSLT 2025 is trying to improve the dubbing task through its open challenge.

Real-time translation requires Simul-S2ST models, which often face a latency-quality tradeoff challenge (Dugan et al., 2023). Developing more effective simultaneous policies is essential to better managing this tradeoff. Additionally, these models must support one-to-many and many-to-one translation within a single framework to accommodate diverse linguistic communities. Capturing speech across various accents, in noisy environments, and with varying microphone placements presents additional challenges for direct S2ST systems. Voice preservation is crucial in this context; however, addressing the potential threat of voice cloning remains a significant challenge to prevent misuse (Jia et al., 2019b, 2022b).

In this era of DL, model sizes are rapidly increasing. However, mobile devices are constrained by limited storage, computing power, and energy capacity. Running models on such devices without internet connectivity requires fully on-device applications. This creates a need for highly efficient models that fit these hardware limitations. There are two possible approaches: design small-sized models, though they often suffer from suboptimal performance, and use larger, high-performing models combined with model compression techniques. However, achieving effective compression while maintaining low latency and high-quality output remains a major challenge (Biadsy et al., 2019; Chen et al., 2021; Rybakov et al., 2023a; Agranovich et al., 2024).

10. Research challenges and future directions

This section highlights key challenges that researchers working on S2ST may explore. Based on the nature of these challenges and potential future directions, they are grouped into three thematic clusters: Data and Evaluation, Deployment Issues, and Model Architectures.

10.1. Data and evaluation

Table 1 presents the available parallel speech data, which provides minimal language coverage compared to the number of languages spoken worldwide. Moreover, the available parallel speech is only semantically aligned, meaning that acoustic features in the source language, such as speaking style, emotion, rhythm, and ambient noise, are not necessarily preserved in the target speech. These acoustic elements are essential for voice preservation. Due to the nonalignment of acoustic information, additional efforts and techniques are required to achieve voice preservation. There is a need for time to develop datasets with broader language coverage that are both semantically and acoustically aligned.

Another important aspect of data collection for multilingual speakers is *code-mixing*, where multiple languages are used within a single sentence or utterance. The current literature reveals a research gap in applying S2ST models to *code-mixed data*. Code-mix data presents challenges like diverse lexicons, syntax variations, and a lack of labeled data. Hence, exploring the following questions would be intriguing: (a) Developing Code-mix S2ST datasets encompassing additional languages, (b) Evaluating the performance of current S2ST models on Code-mix S2ST data, and (c) Investigating whether pre-training across multiple languages aids in addressing code-mixing challenges.

As discussed over *Cascade and Direct S2ST Models* in Sections 6.1.1 and 6.1.2, there has been limited empirical study comparing the cascade and direct S2ST models. Furthermore, to our knowledge, no thorough assessment has been done for low-resource languages using direct and cascade models. It may be interesting to compare direct and cascade S2ST models on various S2ST datasets, particularly for low-resource and unwritten languages. Although *error propagation* is a well-known issue in cascade S2ST models, there is a lack of thorough analysis quantifying its impact on cascade models and evaluating the extent to which direct models are immune to error propagation.

Current S2ST systems are evaluated by conducting ASR on the generated waveform and comparing it to the ground truth target text. Errors in ASR can impede evaluation via BLEU. As emphasized in Marie et al. (2021), the BLEU score is reported by more than 99% of MT papers without considering statistical significance testing or human evaluation. Our examination of S2ST papers indicates a similar trend. The development of metrics that directly compare and evaluate source and target speech utterances without resorting to textual analysis remains an open challenge.

10.2. Deployment issues

In real-world scenarios, audio or video often feature multiple speakers in a noisy environment, each potentially with their accent, dialect, pitch, and tone. Conducting *speech separation* beforehand could prove beneficial before inputting the data into the S2ST. Similarly, factors such as ambient noise, background music, cross-talk, and non-verbal sounds can pose challenges for S2ST model training. Distinguishing between meaningful speech and ambient noise presents a non-trivial task for the model.

The recent surge in interest in *multilingual S2ST* is driven by its significance in real-world applications. For instance, a single speech may need to be delivered to multilingual communities, such as during a conference attended by a diverse audience. Multilingual S2ST can encompass various scenarios, including one-to-many, many-to-one, and many-to-many language translations. However, our literature review indicates a scarcity of research in this area. Moreover, there is an

opportunity to explore *simultaneous multilingual S2ST*,[4] representing the most practical setting.

The true challenge for direct models lies in attaining both lower latency and higher translation quality to fulfill real-time usage requirements. Simultaneously preserving the authenticity of the voice presents a challenge, as it is essential to guard against the misuse of *voice cloning*. A hybrid model design combining discrete units and spectrograms can acquire linguistic and para-linguistic features, thereby enhancing translation quality, naturalness, and voice preservation.

Offline direct S2ST models find applications in the dubbing of movies and lectures. Several things must be considered when deploying direct S2ST models for such applications. Firstly, getting a clean speech from a real-world environment is challenging. Cross-talk, noise, and background music removal must be done. Secondly, the challenge lies in handling various accents while translating, and voice preservation should be taken care of. The third important requirement of dubbing is *isochrony*, i.e., dubbed speech has to closely match the timing of speech and pauses of the original audio. Inserting pauses, handling verbosity, and prosodic alignment are a few techniques to achieve isochrony. On the other hand, Simul-S2ST is more practical for real-world dubbing. Contrary to offline S2ST, these models require latency-quality handled in a better way along with isochrony. Developing on-device applications with limited memory and computational power presents a significant challenge, making model compression while maintaining the same latency and quality a particularly demanding task (Agranovich et al., 2024; Rybakov et al., 2023a; Biadsy et al., 2019; Chen et al., 2021).

10.3. Model architecture

Most current research efforts have concentrated on constructing S2ST models and datasets for languages with ample resources. However, given that the effectiveness of S2ST models hinges on parallel speech-to-speech corpora, thus, there is merit in constructing models that can transfer learning from language pairs with abundant resources to those with limited resources.

Many direct models primarily rely on autoregressive models as their sub-modules, in which the current output depends on previous inputs, resulting in a high degree of input dependency. Decreasing the number of modules or layers is essential to reduce the average latency of direct S2ST models. Moreover, shifting the focus from autoregressive models to non-autoregressive models is advisable. This shift in focus is crucial for enabling real-time usage scenarios. LLM-based models have been explored for multilingual and zero-shot S2ST; however, further efforts are required to extend their capabilities to low-resource languages. It would also be interesting to investigate their performance in the context of simultaneous S2ST.

11. Conclusion

We present a comprehensive survey of direct S2ST models. In particular, we discussed the comparison of direct and cascade models, data scarcity, representation, and segmentation issues in S2ST modeling. A bird-eye view of architectures of offline, simultaneous, and LLM-based S2ST models is discussed in detail. We also offer training strategies and application issues that may be beneficial to industry practitioners. Finally, we enumerate a list of open problems and challenges that remain to be solved. Thus, we hope that the review will be a torch-bearer for someone starting their work in the S2ST domain.

CRediT authorship contribution statement

Mahendra Gupta: Writing – original draft. **Maitreyee Dutta:** Supervision. **Chandresh Kumar Maurya:** Writing – review & editing.

[4] In May 2024, Microsoft launched such a service in their Azure product.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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